**Queen Mary University of London**

School of Economics and Finance

Empirical Dissertation

# Deep Hedging SPX Straddles: Simulation Transfer, Market Fine-tuning, and Maturity Pooling

**Name and Surname:** Lewis Hikari Kawase Kennedy

**Module Code:** ECOM107

**MSc Programme:** MSc Finance and Machine Learning

**Name of Supervisor:** Dr Katerina Panopoulou

# Abstract

This dissertation studies whether neural networks improve daily hedging of short, near-at-the-money SPX straddles when the hedge is discrete and incurs transaction costs. It addresses three questions. First, does a learned hedge change the distribution of terminal hedging outcomes relative to conventional delta hedges? Second, do simulation pretraining, fine-tuning on market data, and richer volatility models provide useful information for hedging? Third, does sharing information across option maturities improve performance when historical samples are limited?

Policies are evaluated out-of-sample on nine SPX market years using only observable information available before execution, with hedges executed at the next close. The results separate the economic impact of deep hedging on risk, the value of simulation transfer, and the effect of maturity pooling.

First, the leading policy, trained solely on Lifted Heston simulations and shared across maturities, records premium-normalised MSE of 0.154984 at one-basis-point proportional stock trading cost. This is 15.9% and 16.3% below vendor delta and Black–Scholes delta, respectively. The reduction comes from lower variance and fewer extreme squared errors, while average dollar P&L, turnover, and realised trading costs worsen. The policy therefore controls the dispersion and tail of hedging outcomes rather than increasing expected short-volatility returns. Second, simulation pretraining produces the strongest observed performance: Heston and Lifted Heston policies outperform market-only training on average, while subsequent market fine-tuning adds negligible value and Lifted Heston does not clearly improve on classical Heston. Third, models that share parameters across maturities outperform a model trained only on one-month options on average. This is consistent with the established multi-task-learning view that parameter sharing can reduce estimation error with scarce data. The findings identify a promising risk-control policy, but do not establish a statistically confirmed neural advantage, a distinct benefit from rough-volatility dynamics, or a deployable trading strategy.

**Keywords:** deep hedging; SPX straddles; Black–Scholes; Heston; Lifted Heston; transfer learning; multi-task learning; walk-forward evaluation; transaction costs.

# Contents

**Abstract** **2**

**List of abbreviations** **5**

**1 Introduction** **6**

1.1 Motivation and research problem . . . . . 6

1.2 Delta-hedged SPX straddles . . . . . 6

1.3 Research questions . . . . . 7

1.4 Contribution and scope . . . . . 7

**2 Literature Review** **8**

2.1 Short-volatility exposure and delta hedging . . . . . 8

2.2 Discrete hedging and transaction costs . . . . . 9

2.3 Deep hedging . . . . . 9

2.4 Stochastic, rough, and Lifted Heston volatility models . . . . . 10

2.5 Maturity pooling . . . . . 11

2.6 Synthesis and empirical gap . . . . . 11

**3 Methodology** **12**

3.1 Experimental design . . . . . 12

3.2 Information, execution, and accounting . . . . . 12

3.3 Neural hedging, simulation, and transfer learning . . . . . 14

3.4 Benchmarks and walk-forward evaluation . . . . . 16

3.5 Evaluation and inference . . . . . 17

**4 Data** **18**

4.1 Data sources and market-state construction . . . . . 18

4.2 Quote filtering and episode construction . . . . . 19

4.3 Calibration and simulated training data . . . . . 20

4.4 Final empirical sample . . . . . 22

**5 Empirical results** **23**

5.1 Hedging risk and economic outcomes . . . . . 23

5.2 Simulation transfer . . . . . 27

5.3 Maturity pooling . . . . . 28

**6 Discussion** **29**

6.1 What the quadratic objective buys . . . . . 29

6.2 Simulation as a policy prior . . . . . 30

6.3 Pooling as cross-maturity shrinkage . . . . . 32

6.4 Limitations and next tests . . . . . 32

**7 Conclusion** **33**

**References** **34**

**A Reproducibility details** **36**

A.1 Sample and numerical checks . . . . . 36

A.2 Data coverage . . . . . 37

A.3 Calibration illustrations . . . . . 38

A.4 Network and inputs . . . . . 41

A.5 Transaction-cost and chronology specification . . . . . 41

A.6 Supporting derivations . . . . . 42

**B Additional robustness analyses** **43**

B.1 Planned comparisons . . . . . 43

B.2 Temporal and regime stability . . . . . 45

B.3 Training and optimisation . . . . . 45

B.4 No-trade benchmark . . . . . 46

B.5 P&L decomposition . . . . . 46

B.6 Maturity, costs, and scale . . . . . 47

# List of abbreviations

| | |
|---|---|
| **ATM** | at the money |
| **bp** | basis point |
| **BS** | Black–Scholes |
| **CRSP** | Center for Research in Security Prices |
| **CVaR** | conditional value-at-risk |
| **DTE** | days to expiry |
| **ES** | expected shortfall |
| **IV** | implied volatility |
| **L2-SP** | L2 starting-point regularisation |
| **MAE** | mean absolute error |
| **MLP** | multilayer perceptron |
| **MSE** | mean squared error |
| **OTC** | over-the-counter |
| **P&L** | profit and loss |
| **ReLU** | rectified linear unit |
| **RMSE** | root mean squared error |
| **RQ** | research question |
| **RV** | realised volatility |
| **SPX** | S&P 500 Index and its listed option class |
| **SPY** | SPDR S&P 500 ETF Trust |
| **USD** | United States dollar |
| **VIX** | Cboe Volatility Index |

# 1 Introduction

## 1.1 Motivation and research problem

Equity-index options often price variance above the variance subsequently realised, thus motivating short-volatility positions. A short near-ATM SPX straddle, however, remains exposed to large moves in either direction. Its call and put deltas may leave the straddle close to delta-neutral at inception, but negative gamma causes the position to become directionally exposed as spot changes. Earning any volatility premium while limiting uncontrolled directional exposure therefore requires dynamic hedging. The resulting problem is how residual risk should be managed once hedging is discrete and costly.

Unlike Black–Scholes delta, which only provides a local hedge, deep hedging learns a dynamic hedging strategy by minimising terminal risk under discrete rebalancing and transaction costs. To evaluate deep hedging models, this study hedges one short straddle at a time using the daily SPX close as a broad-market hedge proxy. The central question, therefore, is whether neural policies reduce out-of-sample terminal hedging risk for short near-ATM SPX straddles relative to conventional delta benchmarks under these constraints, and which elements of the training design account for any improvement.

## 1.2 Delta-hedged SPX straddles

The choice of near-ATM straddle is deliberate. It concentrates initial gamma, vega, and volatility exposure around the prevailing index level rather than across the full implied-volatility surface. As spot and time to expiry evolve, rebalancing changes the stock position, incurs transaction costs, and generates path-dependent trading gains or losses. Hedge timing and inventory management are therefore central to the natural sequential-control problem in which a learned policy can be compared directly with conventional delta hedging. This distinguishes the straddle from other short-volatility implementations.

Alternative short-volatility instruments serve different purposes. A short variance swap targets the difference between a predetermined variance strike and subsequently realised variance, providing relatively direct exposure to the realised-variance payoff with limited dependence on the ordering of returns. Its option-strip replication, however, spans a range of strikes, so the fair variance strike incorporates the full implied-volatility surface. This includes the elevated prices of downside options, so a short variance swap also captures compensation embedded in downside skew and tail insurance rather than concentrating the initial exposure around ATM. On the other hand, VIX futures reference future implied volatility, and thus introduce term-structure and convergence risk. By extension, they express related short-volatility views through materially different payoff, skew, term-structure, path, and implementation exposures.

The instruments also differ operationally; variance swaps are OTC, while VIX futures and SPX options are exchange-traded. SPX is particularly suitable for a systematic index-volatility study because its listed market provides observable market quotes and a long, consistent historical panel of option prices, implied volatilities, and benchmark sensitivities. This contrasts with OTC variance swaps, for which comparable historical transaction and quote data are much less accessible. Furthermore, SPX provides broad S&P 500 exposure with European exercise and cash settlement. This eliminates early assignment and the possibility of receiving or delivering an unwanted share position at expiry, simplifying the management of multi-leg

short-option books (Cboe Global Markets, 2026). These features allow the option universe to be held fixed while analysis concentrates on the hedging policy.

## 1.3 Research questions

This setting highlights three related empirical gaps that remain unresolved in existing literature. First, it remains unclear how a learned hedge transforms the distribution of terminal outcomes relative to conventional delta benchmarks. Second, existing evidence does not clearly separate the value of simulation pretraining from subsequent market fine-tuning or from richer simulated volatility dynamics. Third, when historical option samples are limited, it remains unclear whether sharing information across maturities reduces estimation error enough to offset genuine differences in gamma, time decay, and rebalancing horizon.

The formal research questions for this dissertation are as follows:

**RQ1 — Hedging performance and economic effects.** Do neural policies using only market-observable inputs reduce out-of-sample terminal hedging risk relative to conventional delta benchmarks and, if so, which part of the terminal P&L distribution changes?

**RQ2 — Simulation transfer and richer volatility models.** Relative to training on market data alone, do simulation pretraining, subsequent fine-tuning, and richer volatility dynamics provide additional transferable hedging information?

**RQ3 — Sharing information across maturities.** Does sharing parameters across entry maturities improve matched one-month performance relative to a one-month-specific model, enough to offset genuine differences in gamma, time decay, and the number of rebalancing opportunities?

## 1.4 Contribution and scope

Together, these questions examine the source of hedging information, the value of sharing scarce data, and the risks reduced.

The conclusions are specific to dynamically hedged SPX straddles and should not be interpreted as evidence for variance swaps, VIX futures, or short-volatility strategies generally. The study evaluates hedging risk conditional on writing the straddle rather than estimating the variance risk premium or maximising expected strategy return. Because the daily SPX close is used as a broad-market hedge proxy, the results also do not constitute a directly deployable trading strategy.

The dissertation proceeds as follows. Chapter 2 develops the economic and literature basis for short-volatility hedging, simulation transfer, and maturity sharing. Chapter 3 formalises the hedging problem, neural policies, benchmarks, and evaluation framework, while Chapter 4 describes the market and simulated data and constructs the empirical sample. Chapter 5 reports the results for the three research questions. Chapter 6 interprets the findings, their limitations, and implications, and Chapter 7 concludes.

# 2 Literature Review

## 2.1 Short-volatility exposure and delta hedging

Let $C_t$ and $P_t$ denote the call and put values for a common strike and expiry, and let $S_t$ be the SPX level. A short straddle is $-C_t - P_t$. If the stock position is set to the call-plus-put delta, $h_t = \Delta_t^C + \Delta_t^P$, the portfolio is locally delta-neutral:

$$\Pi_t^{\Delta} = -(C_t + P_t) + h_t S_t, \qquad h_t = \Delta_t^C + \Delta_t^P, \qquad \frac{\partial \Pi_t^{\Delta}}{\partial S_t} = 0. \tag{2.1}$$

Delta is the local sensitivity of option value to SPX, so Equation (2.1) removes first-order spot exposure only at the instant of rehedging. The hedge immediately becomes stale as spot and other state variables move. Gamma measures how delta changes with spot, vega measures sensitivity to implied volatility, and theta measures the effect of time passing. For a short straddle their signs are:

$$\Gamma_t^{\Pi} = -\left(\Gamma_t^C + \Gamma_t^P\right) < 0, \qquad \nu_t^{\Pi} = -\left(\nu_t^C + \nu_t^P\right) < 0, \qquad \Theta_t^{\Pi} = -\left(\Theta_t^C + \Theta_t^P\right) > 0. \tag{2.2}$$

The Greek signs in Equation (2.2) explain why delta hedging does not make the position riskless. Over a short hedge interval, under a smooth diffusion and locally fixed implied volatility, the short position's incremental P&L has the familiar variance-spread approximation

$$d\Pi_t^{\Delta} \approx \frac{1}{2}\left(\Gamma_t^C + \Gamma_t^P\right) S_t^2 \left(\sigma_{imp,t}^2 - \sigma_{real,t}^2\right) dt - dTC_t. \tag{2.3}$$

Equation (2.3) links the delta-hedged position to the variance spread, where $\sigma_{imp,t}^2$ is implied variance, $\sigma_{real,t}^2$ is variance realised over the hedge interval, and $dTC_t$ is trading cost.[1] By suppressing first-order spot exposure, the hedge leaves the short-gamma position primarily exposed to the difference between implied and realised variance: it gains when implied variance exceeds subsequently realised variance and loses when realised variance is higher, net of rehedging costs. Positive theta should therefore be interpreted as carry earned while bearing negative convexity and short-volatility risk, rather than as a riskless time-decay return. This relationship is approximate; discrete rebalancing, jumps, volatility-surface changes, and carry generate residual P&L.

Equity-index options have historically priced variance above the variance subsequently realised, consistent with a volatility risk premium for bearing losses during volatility shocks (Bakshi and Kapadia, 2003; Carr and Wu, 2009). Time decay can contribute to realised P&L when spot and implied volatility remain sufficiently stable, while the persistence and mean reversion of volatility can make elevated volatility following a shock subsequently decline. Profitability nevertheless requires the premium earned to exceed realised-move losses and hedging costs. This dissertation takes the written straddle as given and studies its hedge.

Figure 2.1 shows the unhedged expiry payoff; dynamic delta hedging leaves it unchanged but manages pathwise stock exposure and first-order directional risk.

[1] Appendix Section A.6.1 derives Equation (2.3) from the straddle's Itô expansion and the Black–Scholes PDE.

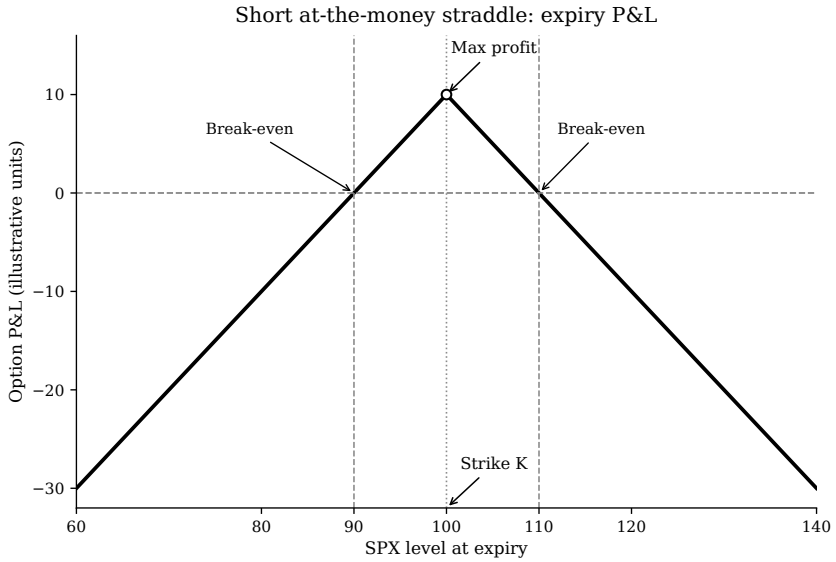



Figure 2.1. Expiry P&L of a short at-the-money straddle. Profit is maximised at the strike and falls beyond the break-even points.

## 2.2 Discrete hedging and transaction costs

The preceding replication argument is the continuous-trading benchmark of Black and Scholes (1973) and Merton (1973). Once trading is discrete or costly, exact replication generally fails. Leland (1985) incorporates rehedging costs through an adjusted-volatility approximation, Hodges and Neuberger (1989) formulate the problem using utility, and Whalley and Wilmott (1997) derive no-trade regions. The common control insight is that the existing hedge position becomes part of the state because changing it is costly.

Rehedging frequency therefore trades tracking error against cost. More frequent rebalancing keeps delta closer to zero and shortens the interval over which negative gamma creates unintended spot exposure, but it raises turnover and transaction costs. Less frequent rebalancing saves cost but leaves a stale hedge through larger spot moves, overnight gaps, and jumps. The hedge is therefore state- and cost-dependent, making Black–Scholes delta a local benchmark rather than a complete policy for a discrete, costly market. A cost-aware controller may rationally tolerate temporary delta exposure when the expected reduction in terminal hedging error is too small to justify another trade.

## 2.3 Deep hedging

Buehler et al. (2019) formulate deep hedging as direct optimisation of a dynamic hedging policy under market frictions, with performance evaluated through the terminal portfolio outcome rather than through a sequence of locally optimal hedge ratios. Transaction costs and trading constraints can enter the optimisation itself, providing the terminal-risk framework adopted here. The closest empirical precedent is Mikkilä and Kanniainen (2023), who apply deep reinforcement learning to S&P 500 options. A policy trained on synthetic data from a calibrated Heston model outperforms Black–Scholes delta in their setting, while direct training on historical intraday data performs even better. Their experiment differs materially from this study: they hedge short-lived calls seven times per day and optimise a risk-return reward, whereas this dissertation hedges near-ATM straddles daily and targets terminal hedging risk. Their result nevertheless makes the training distribution and rehedging clock empirical design choices rather than implementation

details.

The training-data choice involves a trade-off. Simulation can generate many controlled episodes, but the learned policy inherits the simulator's assumptions and estimation error. Historical data contain realised market behaviour without committing to one stochastic law, but offer relatively few independent volatility regimes. Wiese et al. (2019) address scarcity through data-driven market generators, while Buehler et al. (2021) and Limmer and Horvath (2023) explicitly treat distributional misspecification and robustness as part of the hedging problem. The key question is how the training distribution affects the learned hedge.

This study compares market-only training with simulation pretraining and subsequent adaptation to historical data. Output-layer fine-tuning changes only the final mapping from learned features to the hedge; full-network fine-tuning allows the whole representation to adapt. L2-SP regularisation penalises large departures from the pretrained parameters (Li, Grandvalet, and Davoine, 2018).

## 2.4 Stochastic, rough, and Lifted Heston volatility models

Heston (1993) models the index and instantaneous variance as a correlated two-factor diffusion with one principal mean-reversion scale. Rough-volatility models instead use a Volterra kernel to represent volatility dependence across a continuum of horizons (Bayer, Friz, and Gatheral, 2016; Gatheral, Jaisson, and Rosenbaum, 2018). Lifted Heston approximates that kernel with a finite positive sum of exponentials, producing a Markovian factor vector with several decay rates (Abi Jaber, 2019; Abi Jaber and El Euch, 2019).

Under the pricing measure, classical Heston specifies the correlated square-root diffusion

$$\begin{aligned} dS_t &= (r_t - q_t)S_t\,dt + S_t\sqrt{V_t}\,dW_t^S, \\ dV_t &= \kappa(\theta - V_t)\,dt + \xi\sqrt{V_t}\,dW_t^V, \\ d\langle W^S, W^V\rangle_t &= \rho\,dt. \end{aligned} \tag{2.4}$$

In Equation (2.4), $V_t$ is instantaneous variance and mean-reverts at speed $\kappa$ toward long-run level $\theta$; $\xi$ controls volatility-of-variance, while $\rho$ is the correlation between spot and variance shocks. The rate and dividend yield are $r_t$ and $q_t$, and the two Brownian shocks drive spot and variance. Parameters $\kappa$, $\theta$, $\xi$, and $v_0$ are positive, and $\rho$ lies in $[-1, 1]$.

To introduce rougher volatility dependence, the Heston variance equation can be written in Volterra form:

$$\begin{aligned} V_t &= g_0(t) + \int_0^t K_H(t-s)\left[-\kappa V_s\,ds + \xi\sqrt{V_s}\,dW_s^V\right], \\ K_H(u) &= \frac{u^{H-1/2}}{\Gamma(H+1/2)}, \qquad 0 < H < \frac{1}{2}. \end{aligned} \tag{2.5}$$

In the Volterra representation, $g_0(t)$ is a forward-variance input, $K_H$ the fractional kernel, and $\Gamma$ the gamma function. The parameter $H \in (0, 1/2)$ controls roughness; the kernel's singularity places greater weight on recent volatility shocks. Let $\alpha = H + 1/2$.

$$\alpha = H + \frac{1}{2} \in \left(\frac{1}{2}, 1\right), \qquad H \in \left(0, \frac{1}{2}\right). \tag{2.6}$$

Lifted Heston replaces $K_H$ with a positive sum of $N$ exponential kernels, with weight $c_j$ and decay rate $x_j$:

$$K_H(u) \approx K_H^{(N)}(u) = \sum_{j=1}^{N} c_j e^{-x_j u}, \qquad c_j > 0,\ x_j > 0. \tag{2.7}$$

The approximation produces factor convolutions $U_t^j$ and aggregate variance $V_t^{(N)}$:

$$V_t^{(N)} = g_0(t) + \sum_{j=1}^{N} c_j U_t^j, \qquad dU_t^j = \left(-x_j U_t^j - \kappa V_t^{(N)}\right) dt + \xi \sqrt{V_t^{(N)}}\, dW_t^V. \tag{2.8}$$

Following Abi Jaber and El Euch (2019) and Abi Jaber (2019), define $Y_t = (U_t^1, \ldots, U_t^N)$. Its coordinates decay at different rates, while aggregate variance is their weighted projection.

$$g_0(t) + \sum_{j=1}^{N} c_j Y_t^{j,a} = g_0(t) + \sum_{j=1}^{N} c_j Y_t^{j,b} = V_t, \qquad Y_t^a \neq Y_t^b. \tag{2.9}$$

Equations (2.7)–(2.9) replace the fractional kernel with a finite collection of exponential decay rates. The resulting lifted factors record volatility shocks at different horizons, while aggregate variance is only their weighted projection. That projection is many-to-one; the same current variance can arise from different fast- and slow-factor compositions. These latent states shape synthetic prices and surfaces but are not directly observed by a trader, so they are deliberately withheld from the hedging policy. The relevant empirical question is whether richer simulated dynamics generate observable price and surface behaviour that transfers into a better stock hedge. Calibration fit and hedge quality need not agree, and stock-only hedging cannot span all volatility risk (Abi Jaber and Li, 2025).

## 2.5 Maturity pooling

Maturity creates a second bias–variance trade-off. Short- and long-dated straddles differ in gamma, time decay, liquidity, and the number of hedge decisions. Separate policies allow each maturity to specialise but divide an already limited sample. A fully shared model uses all maturities and conditions on original maturity and current time to expiry; an intermediate multi-task model shares hidden features while retaining maturity-specific output layers, following Caruana (1997). The experiment tests whether shared data reduce estimation variance more than common parameters increase bias.

## 2.6 Synthesis and empirical gap

Existing work separately motivates cost-aware hedging, simulation transfer, richer volatility dynamics, and parameter sharing, but does not identify their incremental value under one matched SPX experiment. Three propositions follow. First, minimising terminal hedging risk can reduce dispersion or tail loss without increasing mean P&L because the objective is risk, not return. Second, simulation helps only when the structure it generates transfers through observable market variables; a richer latent volatility model or better static fit need not yield a better stock hedge. Third, parameter sharing should help when maturities share enough hedge structure to offset the bias imposed by common parameters, and hurt when maturity-specific adjustments dominate that sample-efficiency gain. Chapter 3 turns these propositions

into controlled comparisons.

# 3 Methodology

## 3.1 Experimental design

The experiment varies three dimensions while holding the liability and evaluation protocol fixed: training-data source, market adaptation, and parameter sharing across maturities. Synthetic pretraining data are generated from Black–Scholes, Heston, or Lifted Heston models; a pretrained network is then evaluated unchanged, fine-tuned only in its output layer, or fine-tuned throughout the network. The maturity design compares a policy estimated only on one-month entries, a single policy shared across all five entry maturities, and a multi-task policy with shared hidden layers and maturity-specific outputs. This yields 27 simulation-pretrained and three matched market-only specifications that provide the counterfactual.

The liability, observable information, execution lag, transaction-cost model, accounting convention, training budget, and evaluation sample are fixed across variants. A lower loss, therefore, can be attributed to the policy or training design rather than to a different sample, timestamp or accounting rule.[2] Table 3.1 summarises these fixed design choices.

Table 3.1. Experimental design at a glance

| **Component** | Specification fixed before evaluation |
|---|---|
| **Instrument** | Short near-ATM SPX call-put straddle; stock-only hedge; one contract per episode; daily close-to-close execution |
| **Entry maturities** | 1W, 2W, 1M, 2M, and 3M; remaining DTE declines through each episode |
| **Simulation models** | Black–Scholes, Heston, and Lifted Heston; Lifted Heston roughness fixed at $H = 0.35$ |
| **Training route** | Simulation pretraining only; output-layer fine-tuning; full-network fine-tuning; or market-data training only |
| **Maturity sharing** | One-month-specific; fully shared across five maturities; or shared hidden layers with maturity-specific outputs |
| **Evaluation** | Rolling 18-month market window: 12 months estimation, 6 months validation, refit on all 18 months, then 12-month out-of-sample test |
| **Seeds** | Five common optimisation seeds: 20260710–20260714 |
| **Costs** | Observed option bid-ask spreads; USD 4.50 fixed option round trip; stock costs of 1, 5, and 10 bp |
| **Primary outcome** | Premium-normalised terminal mean squared error (MSE) at 1 bp stock cost |

## 3.2 Information, execution, and accounting

Each episode contains one short SPX call-put straddle and a stock hedge. The option position is opened once and is not traded again until close-out; only the stock inventory is rebalanced. Let $t_0 < t_1 < \cdots < t_N$ denote the sequence of daily closes, where $t_0$ is the signal close used to select the contract, $t_1$ is the economic entry close, and $t_N$ is the close-out close. Information stamped no later than $t_i$ forms the decision-time information set. The policy chooses a target stock position $u_i$ at $t_i$, but that target is not executed until the next scheduled close, $t_{i+1}$. If $h_i$ denotes the stock position already held at $t_i$, then $h_0 = 0$ before entry and

[2]The full reproducibility specification is reported in Appendix Table A.1.

$$h_{i+1} = u_i, \qquad i = 0, \ldots, N-2. \tag{3.1}$$

Equation (3.1) is the chronology constraint used everywhere in the study. The option and the first stock hedge therefore enter together at $t_1$, and no new hedge request is created when there is no later holding interval. The one-close delay prevents the same closing observation from being used both to choose and to execute a hedge. Black–Scholes sensitivities are formed from information available at $t_i$, but time to expiry is rolled forward to the scheduled execution close. The price paid for this protection against look-ahead is a genuine overnight exposure.

The simulator index $m \in \{B, H, L\}$ denotes Black–Scholes, Heston, and Lifted Heston, respectively. At node $i$, simulator $m$ maps its internal state $X_i^m$, together with the contract and market variables, into observable synthetic prices and volatility-surface quantities:

$$O_i = g_m(S_i, X_i^m, K, T, r_i, q_i), \qquad m \in \{B, H, L\}. \tag{3.2}$$

The internal state $X_i^m$ is required to generate the synthetic market but is never supplied to the hedging policy. Heston and Lifted Heston can therefore improve the hedge only if their different latent dynamics produce useful differences in observable prices, implied volatilities, or their histories. At each decision close, the policy input $z_i$ is constructed from the contemporaneous observable state $O_i$, causal histories derived from earlier observable states, contract metadata, the current hedge, costs, carry inputs, and missingness indicators:

$$z_i = \left(x_i^{con}, x_i^{surf}, x_i^{hist}, x_i^{ctrl}, x_i^{carry}, \mathbf{M}_i\right). \tag{3.3}$$

Table 3.2 expands the five feature groups and missingness vector in Equation (3.3) and gives notation for every constituent input. A superscript $x$ means that the variable is measured for the scheduled execution date rather than the signal date. Grouping related inputs keeps the state definition readable while preserving an explicit mapping from each feature to its symbol.

Table 3.2. Notation for the observable policy input vector

| Symbol | Meaning | Why it is included |
|---|---|---|
| $x_i^{\text{con}} = (m_i^S, m_i^F, d_i^x, \tau_i^x, d^0, g, v_i^{\text{ctr}}, \Delta_i^{\text{BS}}, \Gamma_i^{\text{BS}})$ | Log spot/forward moneyness; execution DTE in days/years; entry DTE and maturity indicators; contract implied variance; rolled Black–Scholes straddle delta and gamma | Contract location, life-cycle position, and benchmark local risk |
| $x_i^{\text{surf}} = (v_i^{\text{ATM}}, s_i^{\text{term}}, \kappa_i^{\text{term}}, s_i^{\text{skew}})$ | ATM variance at 1W, 2W, 1M, 2M, and 3M, plus surface slope, curvature, and strike skew | Observable volatility level, term structure, and shape |
| $x_i^{\text{hist}} = (r_i^S, RV_i^{(5)}, RV_i^{(20)})$ | Most recent log return and 5-/20-day realised volatility | Causal summaries of recent market behaviour |
| $x_i^{\text{ctrl}} = (h_i, c_i, \Delta t_i^{\text{dec}}, \Delta t_i^{\text{exe}})$ | Current executed hedge, stock-cost rate, time since prior decision, and delay to execution | Inventory, frictions, and irregular calendar gaps |
| $x_i^{\text{carry}} = (r_i, q_i, A_i^r, A_i^q)$ | Interest rate, dividend yield, and age of each carry observation | Financing/dividend carry and data staleness |
| $\mathbf{M}_i$ | Missing-value indicators for derived inputs | Distinguish an unavailable feature from a genuine numerical value |

The previous hedge $h_i$ is part of the state because, with transaction costs, the cost of moving to a new target depends on the position already held. Current DTE measures proximity to expiry; entry maturity identifies the original task, a distinction required for the pooling experiment.

Terminal P&L is computed with one common self-financing accounting engine. The short straddle is sold at its entry bid and repurchased at its close-out ask. Stock trading costs are proportional to traded notional; $F = \text{USD } 4.50/100 = 0.045$ SPX index points is the fixed option round-trip charge for the 100 multiplier. In Equation (3.4), $G^{\text{stock}}$ is the cumulative stock-hedging gain, $FC$ is financing carry, $DC$ is dividend carry, and $TC$ is proportional stock trading cost:

$$\Pi = P^{\text{str}}_{1,\text{bid}} - P^{\text{str}}_{N,\text{ask}} + G^{\text{stock}} + FC + DC - TC - F. \tag{3.4}$$

Equation (3.4) makes the economic components explicit. The first two terms are the short-option cash flows. $G^{\text{stock}}$ is calculated by summing, over each holding interval, the stock inventory held during that interval multiplied by the subsequent SPX price change. $TC$ sums the proportional cost rate times SPX notional times the absolute change in inventory, with zero inventory before entry and after final liquidation so both endpoint trades are charged. The option bid-ask spread enters directly because the straddle is sold at the bid and repurchased at the ask. Financing, dividend carry, and final liquidation are therefore part of the hedge outcome rather than post-hoc adjustments.

Because longer-dated straddles carry larger dollar premia, raw P&L is normalised by the entry straddle mid-price. The terminal outcome $Z$ and the training loss are

$$Z = \frac{\Pi}{P^{\text{str}}_{1,\text{mid}}}, \qquad \mathcal{L}(\theta) = \mathbb{E}[Z(\theta)^2]. \tag{3.5}$$

This follows the terminal-risk deep-hedging framework of Buehler et al. (2019), with MSE as the objective used here. It does not maximise expected return. Squaring the terminal outcome penalises large positive and negative deviations from zero. The standard identity

$$\mathbb{E}\left[Z^2\right] = \text{Var}(Z) + (\mathbb{E}[Z])^2, \tag{3.6}$$

shows why a lower MSE can arise either from lower dispersion or from a mean terminal P&L closer to zero.[3] Mean P&L, MAE, expected shortfall, turnover, and realised costs are therefore reported alongside MSE. Final accounting is recomputed in float64 and required to reconcile to Equation (3.4).

## 3.3 Neural hedging, simulation, and transfer learning

All neural policies use the same modest feed-forward network so that the experiment focuses on information and training design rather than model size. The network has two hidden layers with 16 and 8 ReLU units. Instead of predicting an unconstrained stock position from scratch, it learns a correction $a_i$ to the Black–Scholes straddle delta recalculated for the scheduled execution date:

$$a_i = 0.50 \tanh(f_\theta(z_i)), \qquad u_i = \text{clip}(\Delta_i^{\text{BS,exec}} + a_i, -2, 2). \tag{3.7}$$

[3] Appendix Section A.6.2 derives Equation (3.6) and the corresponding matched-policy MSE contrast.

The hyperbolic tangent bounds the neural correction at $\pm 0.50$, while the final clip bounds the requested stock position at $[-2, 2]$ index units. The output layer is initialised at zero, so $a_i = 0$ before fitting and the neural policy initially reproduces the Black–Scholes benchmark. A correction to a familiar hedge is easier to interpret than an arbitrary position.

Maturity sharing changes which parameters are common across contracts. The one-month-specific model is fitted only to 1M entries. The fully shared model uses all five maturity groups with one common output mapping and relies on current DTE and original maturity to distinguish them. Following Caruana (1997), the multi-task model shares its hidden representation but gives each maturity group (g) a separate linear output head:

$$f_{\theta,g}(z_i) = w_g^\top \phi_\psi(z_i) + b_g. \tag{3.8}$$

In Equation (3.8), $\psi$ contains the shared hidden-layer parameters, while $w_g$ and $b_g$ are the weights and intercept of the output head for maturity group $g$. The three architectures are compared on the same 1M test episodes when maturity pooling is the object of interest. The comparison therefore asks whether the extra data and common structure from other maturities outweigh the bias introduced by forcing some parameters to be shared.

Figure 3.1 makes the architecture comparison explicit: all models receive the same observable inputs and use the same small multilayer-perceptron trunk; only maturity sharing differs.[4]

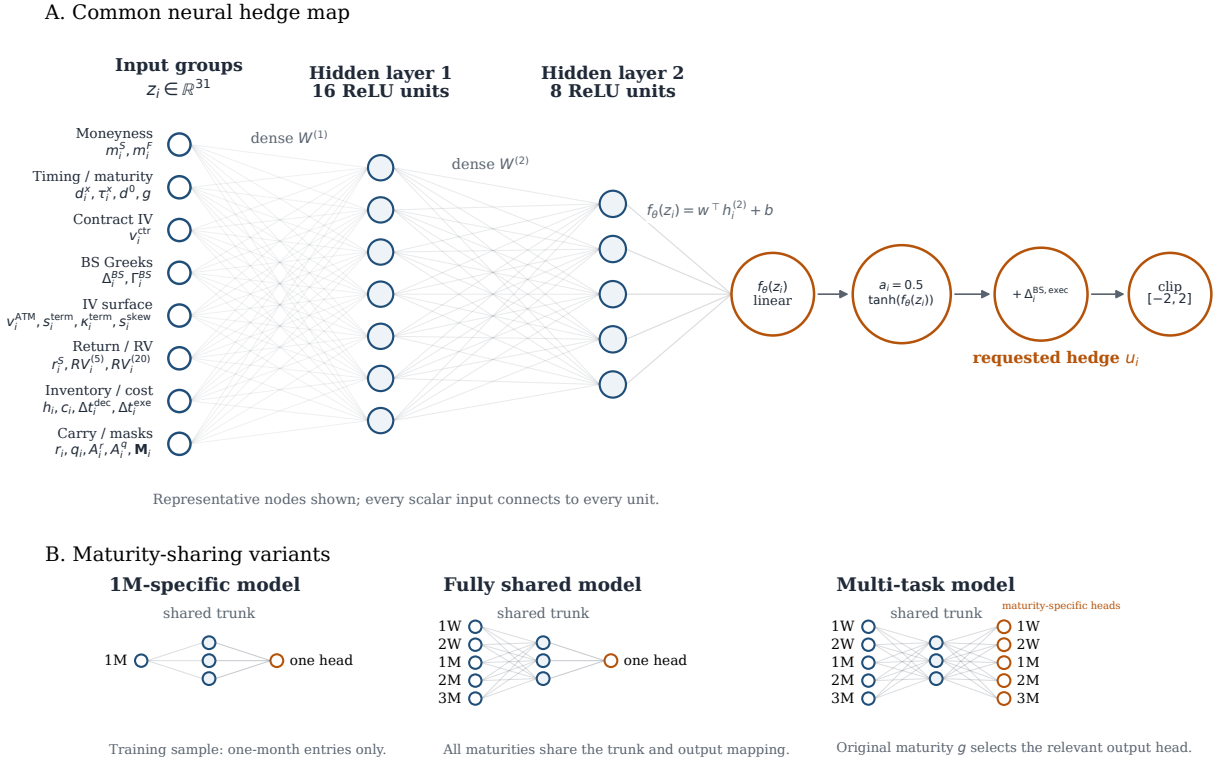



Figure 3.1. Neural policy architecture and maturity-sharing variants. All models use the same observable input and two hidden ReLU layers; they differ only in how parameters are shared across maturities at the output stage.

[4]The parameter counts and output-layer update counts for each architecture are reported in Appendix Table A.5.

The simulation stage uses the Black–Scholes, Heston, and Lifted Heston models described in Chapter 2. For Lifted Heston the experiment fixes $H = 0.35$, $\alpha = H + 1/2 = 0.85$, $N = 20$ exponential factors, and geometric-grid ratio $r_N = 2.5$ before downstream evaluation (Abi Jaber, 2019; Abi Jaber and El Euch, 2019). Because $H$ is fixed before the hedging experiment, the results do not identify a causal effect of roughness. The Heston Feller condition $2\kappa\theta \geq \xi^2$, a sufficient condition for strictly positive variance in the continuous model, is not imposed on every calibration date; the numerical scheme enforces non-negative simulated variance.

Training is episode-based: one observation for the optimiser is a complete option liability and all of its hedge decisions, not a single daily action. For a mini-batch containing $B$ complete episodes, simulation pretraining and market-only training minimise the sample analogue of the terminal MSE in Equation (3.5):

$$\widehat{\mathcal{L}}_B(\theta) = \frac{1}{B}\sum_{e=1}^{B} Z_e(\theta)^2. \tag{3.9}$$

A simulation-pretrained policy is then evaluated unchanged or adapted using the preceding market window. Output-layer fine-tuning minimises the same loss but updates only the final output parameters. Full-network fine-tuning updates every layer and adds L2-SP regularisation (Li, Grandvalet, and Davoine, 2018):

$$\widehat{\mathcal{L}}_{\mathrm{FT}}(\theta) = \frac{1}{B}\sum_{e=1}^{B} Z_e(\theta)^2 + \lambda_{\mathrm{SP}}\|\theta - \theta^{\mathrm{pre}}\|_2^2. \tag{3.10}$$

In Equation (3.10), $\theta^{\mathrm{pre}}$ is the parameter vector learned during simulation pretraining and $\lambda_{\mathrm{SP}}$ is the regularisation weight. The penalty discourages the market adaptation step from discarding the representation acquired in simulation unless the historical data provide enough evidence to move it. Policies are estimated with AdamW, gradient clipping, and fixed training budgets. No future quote, later calibration, interest-rate observation, or dividend observation enters a decision.

## 3.4 Benchmarks and walk-forward evaluation

Neural policies are evaluated against conventional delta targets under the same information, timing, units, and accounting. The primary classical benchmark is the Black–Scholes call-plus-put delta recalculated for the scheduled execution date. Additional benchmarks use the OptionMetrics vendor call-plus-put delta, a Heston delta from the most recent admissible calibration, and a reset-state Lifted Heston delta from the most recent admissible calibration. Therefore, each benchmark faces the same one-close delay, transaction costs, and carry assumptions as the neural policies.[5]

Because continuously trading to a point delta is generally not optimal when trading is costly, each benchmark delta is also paired with a state-dependent no-trade rule. Equation (3.11) compares the currently held stock inventory with a band around the benchmark target:

$$u_i^{\mathrm{NT}} = \mathrm{clip}(h_i, \delta_i - w_i, \delta_i + w_i). \tag{3.11}$$

[5]The raw benchmark targets and their no-trade variants are tabulated in Appendix Table A.6; validation-selected no-trade scales are shown in Appendix Figure B.4.

The rule does nothing while the current inventory lies inside the band. If the inventory lies outside it, the position moves only to the nearest boundary rather than all the way to the point-delta target. The width follows the cube-root transaction-cost scaling motivated by Whalley and Wilmott (1997):

$$w_i = \min\left\{0.5,\ a\left(c_i S_i |\Gamma_i^{\mathrm{BS}}|^2 (\sigma_i^{\mathrm{IV}})^2 \Delta t_i\right)^{1/3}\right\}. \tag{3.12}$$

The band is capped at 0.5 index units. Equation (3.12) widens the band with higher proportional stock costs, spot levels, absolute Black–Scholes gamma, implied volatility, or holding intervals. The scale parameter $a$ is selected only on the validation sample inside each walk-forward window.

Evaluation is strictly walk-forward. For each test year, the preceding 18 months are split into 12 months for initial estimation and 6 months for validation. Training budgets and no-trade-band scales are chosen using that validation period. The selected procedure is then refitted on the full 18-month pre-test window and evaluated during the following 12 months. This produces nine annual out-of-sample periods from 2016 to 2024 with all five maturity groups represented. Separating model selection from final evaluation follows the general principle that reuse of test data creates optimistic performance estimates (Cawley and Talbot, 2010).

Five common random seeds are used for every neural specification so that optimisation variation can be observed under matched initialisation randomness. Seeds do not create five independent market histories. Instead, they measure training instability, while the realised market year remains the independent economic unit. This distinction is important when neural-network results are based on a small number of genuinely different market regimes (Agarwal et al., 2021).

## 3.5 Evaluation and inference

The primary performance metric is premium-normalised terminal MSE at a one-basis-point stock transaction cost. For a given test year, entry-maturity group, and optimisation seed, Equation (3.13) averages squared terminal outcomes across the matched option episodes:

$$\mathrm{MSE}_{y,g,s}(A) = \frac{1}{n_{y,g}} \sum_{e \in \mathcal{E}_{y,g}} (Z_{e,s}^{A})^2. \tag{3.13}$$

Episodes within a year and maturity are therefore averaged before the broader comparison is formed. For two policies $A$ and $B$, the principal nine-year paired estimand is

$$D(A,B) = \frac{1}{9 n_G n_S} \sum_{y=2016}^{2024} \sum_{g \in G} \sum_{s \in S} \left[\mathrm{MSE}_{y,g,s}(A) - \mathrm{MSE}_{y,g,s}(B)\right]. \tag{3.14}$$

In Equation (3.14), $G$ is the set of maturity groups in the comparison, $S$ is the common seed set, and $n_G$ and $n_S$ are their respective counts. For the all-maturity analysis, $G$ contains all five entry maturities; for the matched maturity-pooling comparison it contains only 1M. Negative $D(A,B)$ favours policy $A$ because it has the lower MSE. For inference, the maturity-and-seed average inside each year is retained as the year-level paired difference $d_y$. Equal weighting across maturity groups prevents longer or more numerous maturity samples from mechanically dominating the result, while equal weighting across the

nine years keeps the inferential unit aligned with independent market periods rather than repeated hedge decisions.

Uncertainty is assessed at the annual level. Nominal intervals use 5,000 bootstrap resamples of the nine year-level differences $d_y$; sign-randomisation tests assess whether the paired annual differences are centred on zero; and Holm adjustment controls family-wise error across the pre-specified comparison set (Holm, 1979).[6] Optimisation-seed dispersion is reported separately rather than treated as extra market evidence.

RMSE, MAE, and mean bias describe the scale and direction of terminal errors. ES95 is reported as a positive loss magnitude: it is the negative of the average terminal P&L outcome in the worst 5% of the distribution. Turnover and itemised realised costs show how much trading is required to obtain any reduction in terminal risk. Results at 5 and 10 bp, by maturity, by year, under leave-one-year-out calculations, and across market regimes are robustness analyses rather than alternative selection rules.

# 4 Data

## 4.1 Data sources and market-state construction

OptionMetrics IvyDB US supplies SPX expiries, strikes, closing bid–ask quotes, implied volatilities, vendor Greeks, rates, and dividend inputs; CRSP (2026) supplies the daily SPX closes used as the common signal and execution proxy. IvyDB US is an end-of-day option database (OptionMetrics, 2026). The market and calibration files cover 2010–2024; Heston and Lifted Heston calibration objects are produced before the hedging experiment as described in Section 4.3.

Every feature supplied to a policy is constructed causally. Equation (4.1) constructs the forward index level from the decision-close spot, interest rate, dividend yield, and remaining maturity, and then defines spot and forward moneyness:

$$F_i = S_i e^{(r_i - q_i)\tau_i}, \qquad m_i^S = \log(S_i/K), \qquad m_i^F = \log(F_i/K). \tag{4.1}$$

Moneyness is therefore zero when spot or forward equals the strike and changes sign depending on which side of the strike the index lies. The forward version removes the mechanical effect of interest rates and dividends over the remaining life of the option.

The benchmark sensitivities are recomputed rather than copied from a future execution observation. Using the contract implied volatility observed at the decision close and the time remaining at the scheduled execution close, the Black–Scholes-Merton call and put formulas imply the following call-plus-put straddle delta and gamma (Black and Scholes, 1973; Merton, 1973):

$$\begin{aligned} d_{1,i} &= \frac{\log(S_i/K) + (r_i - q_i + \frac{1}{2}\sigma_i^2)\tau_i^x}{\sigma_i\sqrt{\tau_i^x}}, \\ \Delta_i^{BS} &= e^{-q_i\tau_i^x}\big(2\Phi(d_{1,i}) - 1\big), \\ \Gamma_i^{BS} &= \frac{2e^{-q_i\tau_i^x}\,\phi(d_{1,i})}{S_i\sigma_i\sqrt{\tau_i^x}}. \end{aligned} \tag{4.2}$$

[6] The full pre-specified comparison family and its forest plot are retained in Appendix Table B.1 and Figure B.1.

$\Phi$ is the standard normal cumulative distribution function and $\phi$ its density. The straddle delta is the sum of call and put deltas; the straddle gamma is twice the single-leg gamma because a European call and put with the same strike and expiry have the same gamma. Rolling $\tau$ forward while keeping all price, volatility and carry inputs dated at $t_i$ preserves the one-close information rule from Section 3.2.

Volatility enters in two forms. Contract implied variance is simply the square of contract implied volatility. The broader surface state contains ATM implied variance at constant maturities of 1W, 2W, 1M, 2M, and 3M together with slope, curvature, and strike-skew features derived from the contemporaneously available surface. The ATM vector describes the surface level and term structure; slope and curvature describe how that level changes across maturity, while skew records the asymmetry of implied volatility across strikes. These variables are generated from market quotes or, for synthetic data, from the observable option prices produced by Equation (3.2); latent simulator factors never appear as features.

Recent market behaviour is summarised by the previous close-to-close log return and annualised realised volatility over five and twenty trading days:

$$r_i^S = \log(S_i/S_{i-1}), \qquad RV_i^{(k)} = \sqrt{\frac{252}{k}\sum_{j=0}^{k-1}(r_{i-j}^S)^2}, \qquad k \in \{5, 20\}. \tag{4.3}$$

Only returns ending at or before the decision close enter these summaries. Timing variables record the gap since the prior decision and the delay to the scheduled execution close. Interest rates and dividend yields are joined using the most recent observation dated no later than the decision; the age of each carry observation is retained as a feature so that a stale observation is distinguishable from a fresh one. Missing derived inputs are accompanied by explicit masks, and feature scaling is fitted only inside the relevant pre-test training window. No test-period observation is used to define a training scaler.

## 4.2 Quote filtering and episode construction

The quote funnel begins on 4 January 2010. Invalid option types and non-positive strikes are removed. Each retained leg must have a bid of at least 0.05 and an ask no lower than the bid. To remove quotes whose spread is too large to be economically informative while retaining a broad sample, the relative spread is required to satisfy

$$RS_i = \frac{2(A_i - B_i)}{A_i + B_i} \leq 1, \tag{4.4}$$

so the maximum allowed spread is 100% of the quote mid. Observed spreads are not replaced by this filter: the actual bid is still used for short-option entry and the actual ask for close-out in the accounting of Section 3.2.

Calls and puts are paired only when date, expiry, and strike coincide. Near-ATM eligibility is based on absolute log-forward moneyness:

$$-0.12 \leq m_i^F = \log(F_i/K) \leq 0.12. \tag{4.5}$$

One deterministic signal date is selected per week. The pipeline searches separately for 1W, 2W, 1M, 2M,

and 3M entries, targeting 7, 14, 30, 60, and 90 days to expiry within ranges fixed before evaluation. Eligible candidates are ranked first by distance to the target DTE, then by closeness to forward ATM, then by straddle spread, followed by deterministic tie-breaks. This fixes one reproducible liability and prevents ex post option selection.

Contract selection uses the signal close; the option and initial stock hedge enter jointly at the next scheduled close under Equation (3.1). An episode is retained only if both legs have a usable path, the observable surface state can be constructed, and a valid close-out exists. Entry maturity remains fixed through the episode, while execution DTE declines as the contract ages.[7]

## 4.3 Calibration and simulated training data

Figures 4.1 and 4.2 compare the two stochastic-volatility generators under matched diagnostic parameters and starting variance. Heston follows the one-factor square-root diffusion in Equation (2.4); Lifted Heston uses the finite-factor kernel in Equations (2.7)–(2.8) with $H = 0.35$, $N = 20$, and $r_N = 2.5$ (Abi Jaber, 2019; Abi Jaber and El Euch, 2019). The comparison is illustrative and uses the same parameter vector in both models so that differences primarily reflect path dynamics rather than starting conditions.

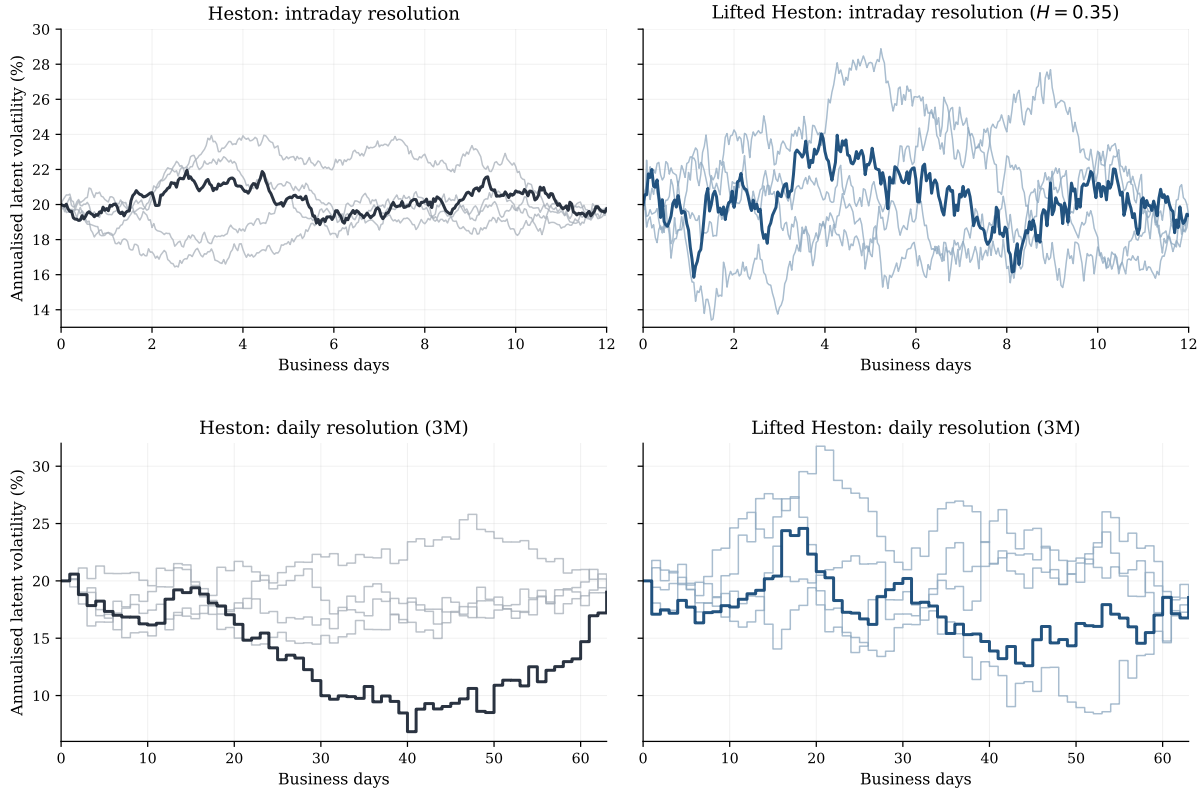



Figure 4.1. Illustrative Heston and Lifted Heston volatility paths. Five paths nearest the median terminal volatility are shown from 10,000 simulations; axes are matched within each row.

The upper panels of Figure 4.1 use an intraday grid over 12 business days, while the lower panels use one simulation step per business day over a 63-day horizon. Only the five paths nearest each model's median terminal volatility are displayed for legibility; the selection is visual and does not enter estimation. Figure 4.2 then uses all 10,000 daily-resolution paths. Lifted Heston generates larger absolute one-day

[7]The minimum sample requirements and the treatment of incomplete test years are documented in Appendix Table A.3.

volatility changes: the mean is 1.20 percentage points versus 0.76 for Heston, and the 95th percentile is 2.94 versus 1.86. Rather than estimating $H$, Figure 4.2 visualises how the two generators differ in one-day volatility variation.

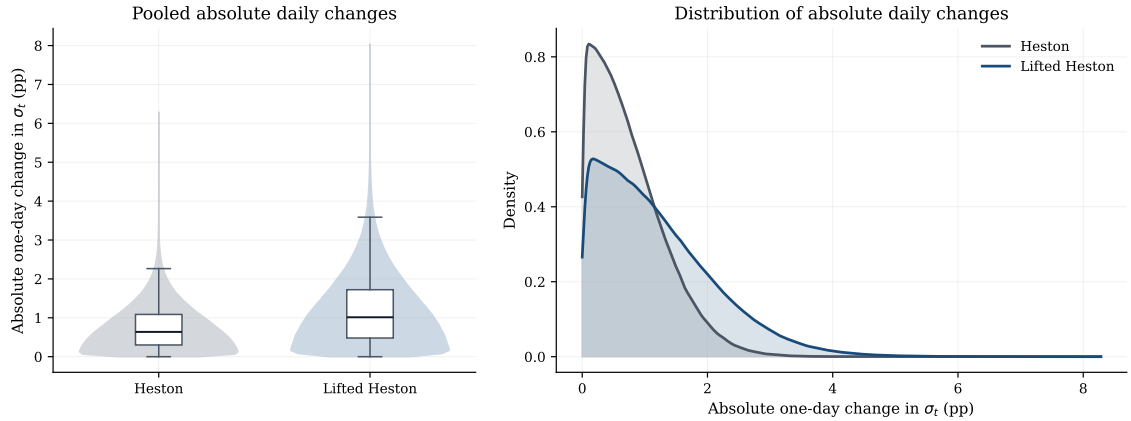



Figure 4.2. Daily roughness diagnostic from 10,000 simulated paths. Lifted Heston produces larger and more dispersed absolute one-day volatility changes than Heston.

The stochastic-volatility inputs used for simulation are fixed before the hedging comparison. A preceding calibration stage produces dated Heston and Lifted Heston parameter panels together with the regularised forward-variance curve. The roughness diagnostic did not identify a single structural value of $H$. Table 4.1 shows the pre-hedging sensitivity analysis used to choose a controlled downstream configuration. Across the aggregate grid, $H = 0.25$ gives the smallest median absolute ATM-skew error, while the vega-scaled fit metric is already relatively flat from approximately $H = 0.25$ onward. In the maturity-pool diagnostic, $H = 0.35$ is selected for both the 1M–3M core pool and the 1W–1M front-end pool, whereas other pools favour values between 0.10 and 0.40. The evidence supports $H = 0.35$ as a fixed experimental compromise, not a universal roughness estimate.

Table 4.1. Pre-hedging Lifted Heston roughness diagnostic used to fix $H = 0.35$

**Panel A: aggregate $H$-grid diagnostic**

| $H$ | Median vega-scaled fit metric | Median absolute ATM-skew error |
|---|---|---|
| 0.10 | 0.007747 | 0.144100 |
| 0.20 | 0.001137 | 0.007584 |
| 0.25 | 0.000951 | **0.005534** |
| 0.30 | 0.000903 | 0.006218 |
| **0.35**† | 0.000868 | 0.006475 |
| 0.40 | 0.000841 | 0.006737 |
| 0.45 | 0.000822 | 0.007001 |

**Panel B: best $H$ when the calibration pool is changed**

| Calibration pool | Included maturities | Best $H$ | Dates | Options | Vega RMSE | Price RMSE |
|---|---|---|---|---|---|---|
| Core | 1M, 2M, and 3M | **0.35** | 30 | 1,621 | 0.008406 | 1.925 |
| Front end | 1W, 2W, and 1M | **0.35** | 30 | 1,305 | 0.020100 | 1.250 |
| All five | 1W–3M | 0.40 | 30 | 1,659 | 0.021000 | 2.599 |
| Excluding 1W | 2W, 1M, 2M, and 3M | 0.30 | 30 | 1,578 | 0.011300 | 2.106 |
| 1M only | 1M | 0.10 | 29 | 1,818 | 0.012800 | 3.138 |
| Mid curve | 2M, 3M | 0.10 | 30 | 1,199 | 0.007485 | 1.085 |
| Very short | 1W, 2W | 0.40 | 14 | 610 | 0.022800 | 1.331 |

*Note:* †*Configuration fixed for the downstream hedging experiment. Panel A demonstrates that the calibration objective and ATM-skew error do not select exactly the same $H$. Panel B shows that the preferred value is maturity-pool dependent. The diagnostic uses option-fit information and is not a search over downstream hedging P&L.*

The maturity dependence is also visible at the fixed $H$: Lifted Heston is closer to market ATM skew at 1M, while Heston is closer at 2M and 3M. The richer simulator therefore improves a short-end surface feature selectively rather than dominating at every maturity.[8] The downstream experiment neither searches over $H$ nor recalibrates it after observing hedge performance.

Synthetic pretraining uses calibration dates in 2010–2011. Each of the Black–Scholes, Heston, and Lifted Heston training sets contains 336 episodes in each of the five entry-maturity groups, giving 1,680 episodes per simulator. Keeping the maturity counts equal prevents a simulator comparison from being confounded by a different mix of short- and long-dated contracts. After simulation, synthetic paths enter the same observable-state and feature-construction pipeline used for market episodes. Simulated and historical episodes therefore share the same variable definitions, units, missing-value masks, and one-close execution chronology.

## 4.4 Final empirical sample

The out-of-sample evaluation contains nine annual test periods from 2016 to 2024, 2,030 distinct option episodes, and 462 weekly entry dates, with all five entry maturities represented in every test year. Earlier years remain available for calibration and rolling estimation but lack sufficiently balanced five-maturity coverage. The 719,370 exported model–episode rows are repeated evaluations of these same liabilities, not independent market episodes.[9] Table 4.2 breaks the sample down by original maturity.

Table 4.2. Evaluation-sample characteristics by original maturity

| Original maturity | Episodes | Entry dates | Median path nodes | Median premium (USD) | Median spread (%) | Median RV20d (%) |
|---|---|---|---|---|---|---|
| 1W | 460 | 460 | 5 | 5,885 | 2.08 | 11.9 |
| 2W | 455 | 455 | 9 | 8,135 | 1.44 | 11.9 |
| 1M | 411 | 411 | 21 | 12,690 | 1.00 | 12.0 |
| 2M | 377 | 377 | 41 | 18,640 | 0.72 | 12.0 |
| 3M | 327 | 327 | 60 | 24,155 | 0.70 | 11.8 |

*Note: Evaluation sample only. Medians are calculated across the 2,030 distinct episodes; path nodes include entry and close-out observations. Source: author calculations from OptionMetrics IvyDB US and CRSP after the filters in Section 4.2.*

The realised sample changes materially with maturity. Median path length rises from five daily nodes for 1W entries to sixty for 3M entries, while median premium rises from about USD 5,885 to USD 24,155. Relative option spreads narrow from 2.08% to 0.70%, whereas entry RV20d remains close to 12% across maturity groups. Longer-dated contracts therefore supply more hedge decisions and tighter proportional option execution, but they also keep the policy exposed for longer and involve larger option premia. These differences are precisely why the maturity-sharing comparison is evaluated on matched 1M liabilities rather than by comparing raw performance across maturity buckets.

[8] The maturity-by-maturity ATM-skew comparison is shown in Appendix Figure A.2. The representative-date diagnostic and six-panel smile slice appear in Appendix Table A.4 and Figure A.3; additional surface, rolling-fit, and forward-variance diagnostics appear in Appendix Figures A.4, A.5, and A.6.

[9] Quote-funnel attrition and maturity coverage are shown in Appendix Figure A.1.

# 5 Empirical results

The results follow the three research questions. Each subsection reports the out-of-sample estimates and highlights the main patterns in the tables and figures.

## 5.1 Hedging risk and economic outcomes

### 5.1.1 Risk and P&L

Table 5.1 reports the Heston, Lifted Heston, and market-trained neural policies alongside the delta benchmarks at the primary one-basis-point stock cost. One-month-specific models enter the matched 1M comparison in Section 5.3.

The ranking is strongly clustered. The three shared Lifted Heston policies occupy the first three positions and differ in MSE by less than 0.001; the Heston and Lifted Heston multi-task triplets are similarly tight. Fine-tuning changes the ordering within clusters, while the simulation model and sharing design create the larger gaps.

Table 5.1. All-maturity performance at one basis point

| Policy basis | Sharing | Training or hedge rule | MSE | RMSE | MAE | ES95 | Bias | Turn. | Cost |
|---|---|---|---|---|---|---|---|---|---|
| Lifted Heston | Shared | Pretraining only | 0.154984 | 0.393680 | 0.270607 | 0.780805 | -0.041287 | 3.573 | 3.842 |
| Lifted Heston | Shared | Output fine-tuned | 0.155185 | 0.393935 | 0.270609 | 0.782292 | -0.042097 | 3.570 | 3.841 |
| Lifted Heston | Shared | Fully fine-tuned | 0.155806 | 0.394722 | 0.270903 | 0.784692 | -0.042645 | 3.563 | 3.838 |
| Heston | Multi-task | Output fine-tuned | 0.159881 | 0.399851 | 0.271466 | 0.799590 | -0.039668 | 3.450 | 3.801 |
| Heston | Multi-task | Fully fine-tuned | 0.160099 | 0.400123 | 0.271702 | 0.800274 | -0.039913 | 3.450 | 3.801 |
| Heston | Multi-task | Pretraining only | 0.160394 | 0.400492 | 0.272161 | 0.799556 | -0.038964 | 3.450 | 3.801 |
| Lifted Heston | Multi-task | Output fine-tuned | 0.163293 | 0.404095 | 0.270569 | 0.816261 | -0.036826 | 3.459 | 3.804 |
| Lifted Heston | Multi-task | Fully fine-tuned | 0.163530 | 0.404388 | 0.270811 | 0.816889 | -0.037128 | 3.458 | 3.804 |
| Lifted Heston | Multi-task | Pretraining only | 0.163918 | 0.404868 | 0.271262 | 0.817259 | -0.036135 | 3.458 | 3.804 |
| Heston | Shared | Fully fine-tuned | 0.167676 | 0.409482 | 0.283685 | 0.832472 | -0.061954 | 3.524 | 3.828 |
| Heston | Shared | Output fine-tuned | 0.168005 | 0.409884 | 0.284283 | 0.831766 | -0.062208 | 3.535 | 3.832 |
| Heston | Shared | Pretraining only | 0.168958 | 0.411045 | 0.285377 | 0.833712 | -0.062220 | 3.543 | 3.835 |
| Market data | Shared | Market training | 0.176529 | 0.420154 | 0.289416 | 0.818296 | -0.020990 | 3.547 | 3.829 |
| Market data | Multi-task | Market training | 0.178146 | 0.422074 | 0.285194 | 0.835885 | -0.027024 | 3.559 | 3.832 |
| Vendor | – | Unadjusted delta | 0.184236 | 0.429228 | 0.285035 | 0.827106 | -0.004305 | 3.250 | 3.733 |
| Black–Scholes | – | Unadjusted delta | 0.185212 | 0.430362 | 0.288433 | 0.840251 | -0.010508 | 3.428 | 3.794 |
| Vendor | – | No-trade delta | 0.185366 | 0.430542 | 0.287866 | 0.826758 | -0.001300 | 3.074 | 3.658 |
| Black–Scholes | – | No-trade delta | 0.185895 | 0.431155 | 0.290417 | 0.839500 | -0.007550 | 3.229 | 3.712 |
| Heston | – | Unadjusted delta | 0.258762 | 0.508687 | 0.363481 | 0.973260 | +0.031610 | 3.498 | 3.815 |
| Heston | – | No-trade delta | 0.265495 | 0.515262 | 0.366988 | 0.973609 | +0.030900 | 3.116 | 3.684 |
| Lifted Heston | – | Unadjusted delta | 0.279766 | 0.528929 | 0.372984 | 1.008518 | +0.028387 | 3.435 | 3.792 |
| Lifted Heston | – | No-trade delta | 0.281754 | 0.530805 | 0.372215 | 1.006659 | +0.022618 | 2.985 | 3.639 |

*Note: Rows are ordered by MSE. Shared models use one network across maturities; multi-task models share hidden layers but use maturity-specific outputs. MSE, RMSE, MAE, ES95, and bias are premium-normalised; ES95 is reported as a positive loss magnitude; cost is mean total realised cost in SPX points. One-month-specific models are excluded from this all-maturity table.*

At one basis point of stock cost, the lowest MSE observed in the evaluated neural grid is produced by the policy pretrained on Lifted Heston simulations, shared across maturities and evaluated before market fine-tuning. The leading policy reduces premium-normalised MSE by 15.9% relative to vendor delta and 16.3% relative to Black–Scholes delta. Expected shortfall also falls, while mean P&L becomes more negative and turnover rises. Both market-trained policies outperform vendor and Black–Scholes delta; however, their turnover is roughly 10% above vendor delta, and their total realised cost is about 3% higher. The conventional Heston and Lifted Heston deltas rank below those four comparators.

Conventional deltas reverse the neural ranking: Heston and Lifted Heston underperform Black–Scholes and vendor delta, although their simulations produce the strongest neural policies. No-trade bands reduce turnover and cost but slightly increase MSE.

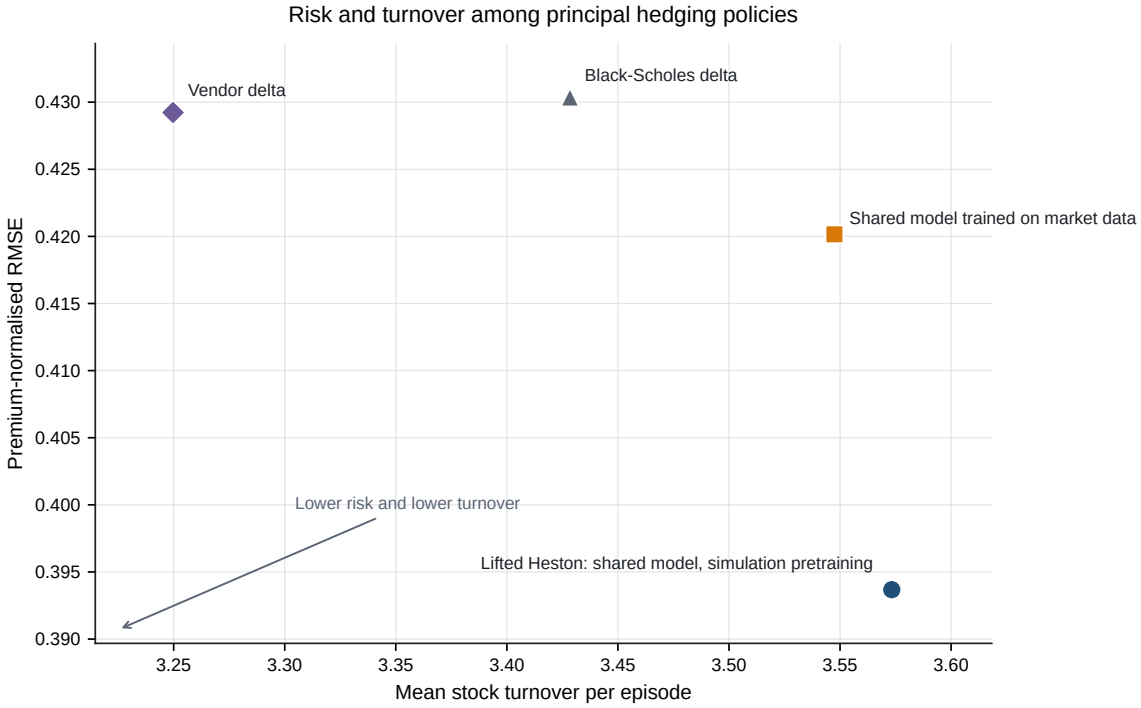



Figure 5.1. Selected points on the risk–turnover frontier at one basis point. Lower and further left are preferable.

Figure 5.1 shows the operational trade-off tracing a downward-sloping frontier: the leading policy reduces dispersion by trading more actively. The policy trained only on market data trades at a similar intensity but removes less risk, which makes it less plausible that the result is merely a consequence of trading more often.

### 5.1.2 Distributional effects

For the pooled episode-level diagnostic, the identity $\mathrm{MSE}(Z) = \mathrm{Var}(Z) + \mathbb{E}[Z]^2$ separates the MSE change into variance and squared-bias components. Relative to Black–Scholes delta, the leading policy reduces variance by 0.030451 but increases squared bias by 0.002023, for a net MSE change of $-0.028428$. Splitting the same difference by the baseline loss distribution assigns $-0.030599$ to the worst 5% of outcomes and $+0.002172$ to the remaining 95%. Almost the entire pooled-sample gain therefore comes from fewer very large squared errors; ordinary outcomes are slightly worse.

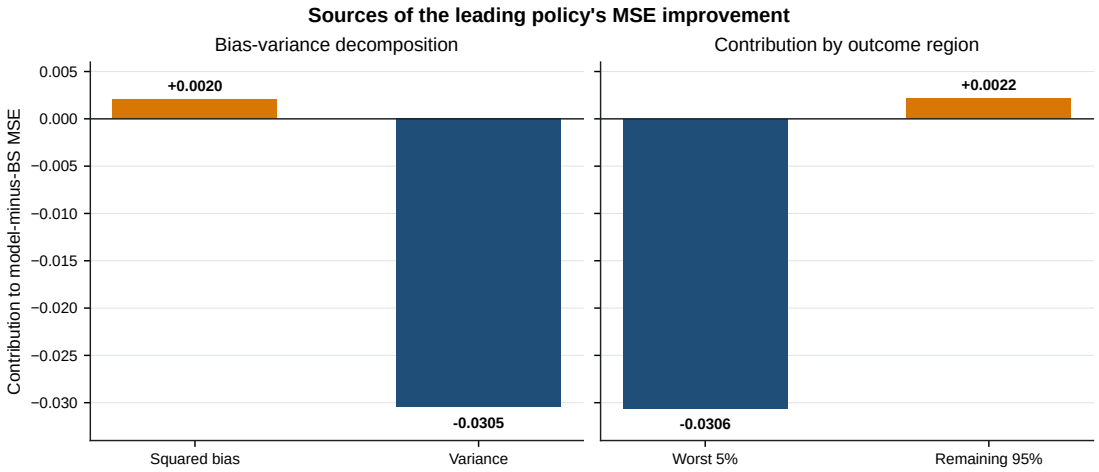



Figure 5.2. Episode-level decomposition of the shared Lifted Heston policy's MSE difference from Black–Scholes delta. Negative contributions favour the neural policy.

The two panels of Figure 5.2 are almost mirror images: the variance reduction is nearly identical to the contribution from the worst 5%, while the squared-bias penalty is nearly identical to the deterioration in the remaining 95%. The figure shows empirically that the observations removed from the far tail are also those driving dispersion, while the centre of the distribution moves slightly away from zero.

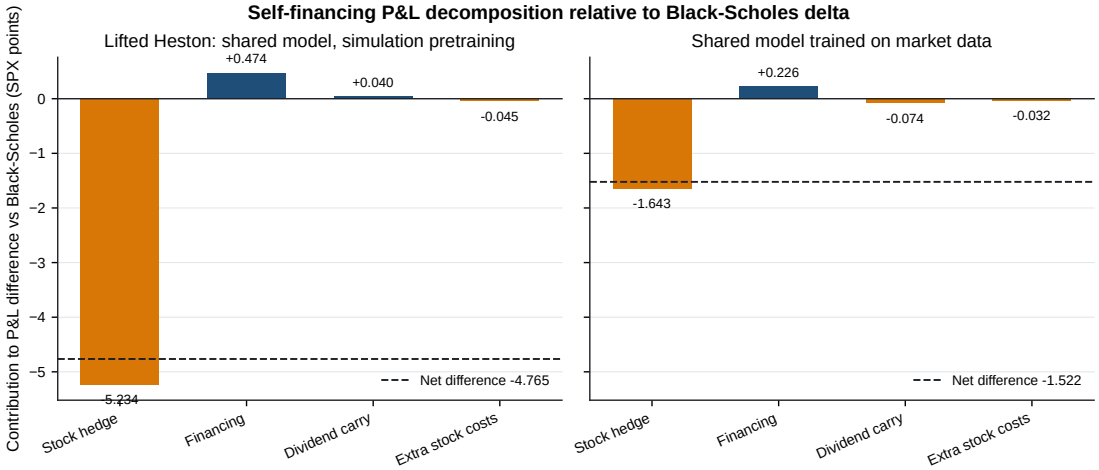



Figure 5.3. Self-financing P&L decomposition relative to Black–Scholes delta. The common option liability cancels from each paired comparison.

Figure 5.3 locates the lower mean in stock-hedging gains. Relative to Black–Scholes delta, the shared Lifted Heston policy loses 5.234 SPX points through stock trading. Financing and dividends recover 0.514 points, while incremental stock cost subtracts 0.045, leaving a net difference of −4.765 points. The market-trained policy has a smaller stock-hedging shortfall of −1.643 points and a net difference of −1.522. Commissions are minor; the stock positions drive the gap.[10]

[10]The component accounting identity is reported as Appendix Equation (B.1).

### 5.1.3 Stability and uncertainty

The annual-weighted MSE difference from Black–Scholes delta is $-0.030228$, with an unadjusted 95% interval of $[-0.066849, 0.000026]$ due to an uneven calendar profile.

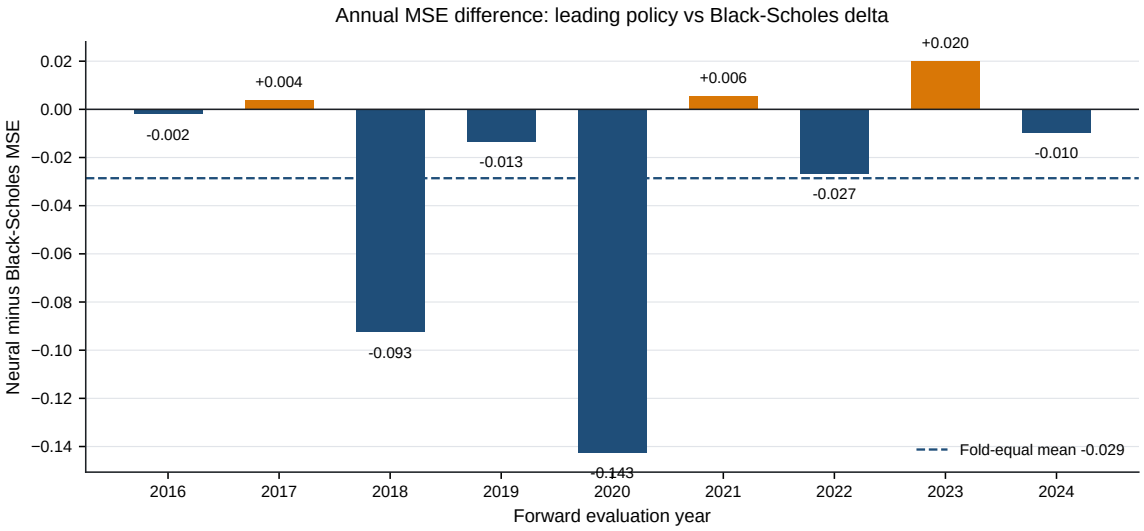



Figure 5.4. Annual paired MSE difference between the shared Lifted Heston policy and Black–Scholes delta at one basis point. Negative values favour the neural policy.

Two deep reductions dominate Figure 5.4: 2018 and 2020 account for 92% of the summed nine-year improvement. The other seven years average only a small gain, and 2023 records the largest underperformance. This profile resembles state-contingent protection, not a uniformly superior delta. Leave-one-year-out differences remain negative, but the economic weight is plainly concentrated.[11]

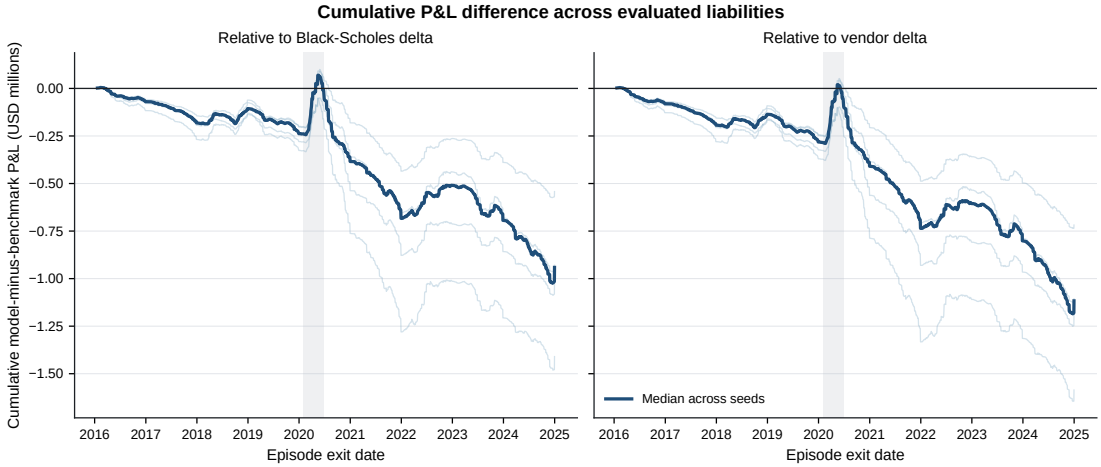



Figure 5.5. Cumulative paired terminal P&L difference of the shared Lifted Heston policy relative to the Black–Scholes and vendor deltas. Pale lines are optimisation seeds; the dark line is their median.

Figure 5.5 adds the pathwise dollar perspective. Cumulative P&L of the leading policy declines through long ordinary stretches, rebounds around some high-error periods, and finishes below zero against both delta benchmarks. The recoveries coincide with the MSE gains in Figure 5.4 but do not offset the persistent stock-hedging shortfall. [12]

[11] Leave-one-year-out and broad annual-RMSE diagnostics appear in Appendix Figures B.2 and B.7.

[12] Appendix Figure B.8 shows the number and aggregate notional of overlapping evaluated liabilities, which is why the

## 5.2 Simulation transfer

### 5.2.1 Pretraining and fine-tuning

Within each Heston or Lifted Heston architecture, the three adaptation variants in Table 5.1 are tightly grouped and have no consistent ordering. Table 5.2 aggregates those matched comparisons.

Table 5.2. Simulation-transfer and model-richness contrasts

| Comparison | MSE difference | Nominal 95% interval |
|---|---|---|
| Output fine-tuning – pretraining only | -0.000442 | [-0.002393, 0.001268] |
| Full-network fine-tuning – pretraining only | -0.000263 | [-0.002745, 0.002085] |
| Simulation pretraining – market-only training | -0.015276 | [-0.041843, 0.002495] |
| Lifted Heston – Heston | -0.005109 | [-0.014990, 0.003047] |

*Note: Negative values favour the first method. Comparisons use Heston and Lifted Heston policies.*

Fine-tuning changes average MSE by less than 0.0005 whether it updates the output layer or the full network. Across the Heston and Lifted Heston policies, simulation pretraining is 0.015276 below market-only training. The performance gap is already present before adaptation to the rolling market window. Lifted Heston is only 0.005109 below Heston on average.

### 5.2.2 Model realism and hedging value

The simulated observable inputs are consistent with the different transfer behaviour. Black–Scholes imposes a flat implied-volatility surface across strikes and therefore cannot reproduce the market's downside skew. Heston and Lifted Heston match skew and term-structure measures more closely, although all three simulated datasets retain material gaps in implied-variance levels and realised volatility. These distributional differences are diagnostics rather than a causal explanation of hedging performance.

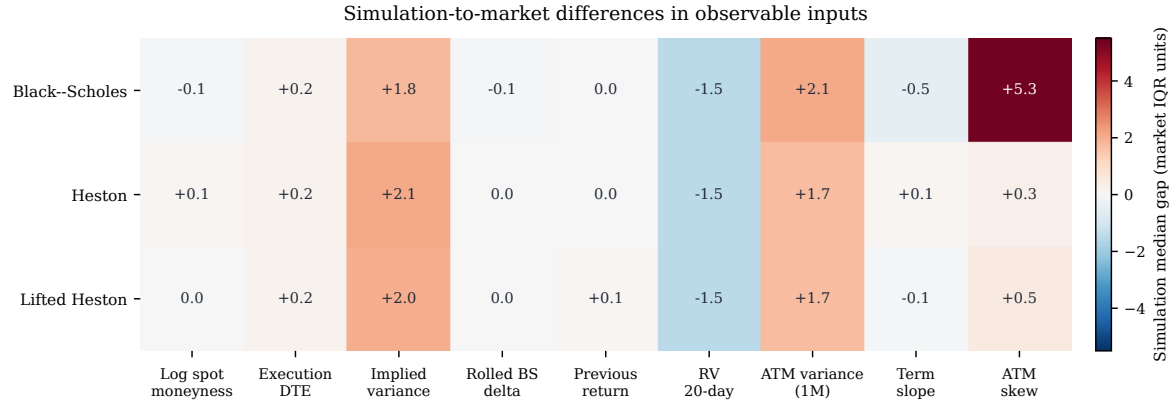



Figure 5.6. Standardised median simulator-to-market differences in observable policy inputs. Values are measured in market interquartile range units.

The heatmap in Figure 5.6 separates into two blocks. Contract-construction variables—log moneyness, time to expiry at execution, execution-horizon Black–Scholes delta, and the previous return—remain close to the market under every simulator. Implied-variance levels, 20-day realised volatility, and one-month ATM variance show sizeable gaps throughout. ATM skew provides the decisive contrast: Black–Scholes is

dollar results should not be read as a small-account backtest.

5.3 market interquartile ranges from the market median, against 0.3 for Heston and 0.5 for Lifted Heston. Despite different latent dynamics, Heston and Lifted Heston generate remarkably similar observable states.

The small and imprecise Lifted-minus-Heston contrast shows that improved surface fit does not automatically improve stock hedging; Section 6.2 develops this interpretation. The weak Heston and Lifted Heston conventional-delta results reinforce the hierarchy: calibration flexibility, local delta accuracy, and terminal hedging performance are separate empirical objects.

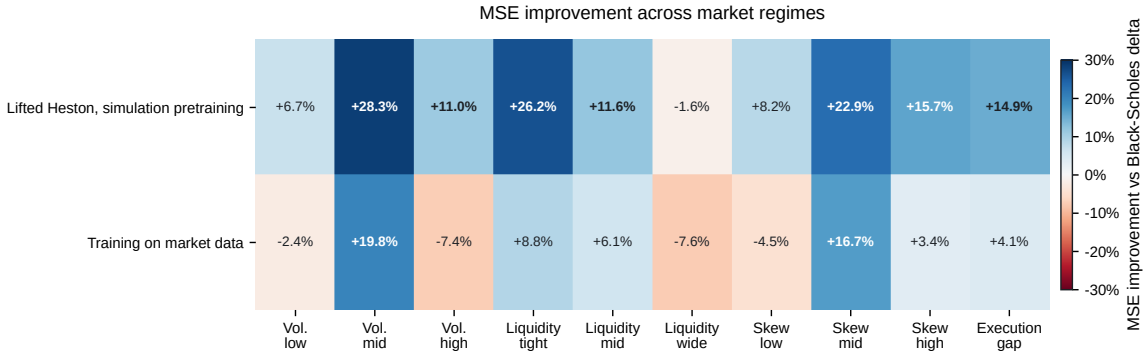



Figure 5.7. MSE improvement relative to Black–Scholes delta across pretest-defined market regimes at one basis point. Positive values favour the learned policy.

Figure 5.7 shifts the comparison from input fit to performance across pretest-defined market regimes. The simulation-pretrained policy improves MSE in nine of the ten regimes, with the largest gains in mid-volatility, tight-liquidity, and mid-skew conditions. Its only reversal is in wide-liquidity conditions. The market-trained policy is less consistent: it performs strongly in the mid-volatility and mid-skew states but falls below Black–Scholes delta at both volatility extremes, in wide liquidity, and under low skew. The two rows peak in similar states, while simulation pretraining extends the improvement across a broader part of the regime grid.

## 5.3 Maturity pooling

Table 5.3 evaluates the three maturity-sharing designs on the same 1M liabilities, making the effect of parameter sharing visible within each training source and variant.

Table 5.3. Complete matched one-month MSE by maturity-sharing design

| Training source | Training variant | 1M-specific | Shared | Multi-task |
|---|---|---|---|---|
| Lifted Heston | Pretraining only | 0.160571 | **0.150951** | 0.159278 |
| Lifted Heston | Output fine-tuned | 0.160614 | **0.151930** | 0.160087 |
| Lifted Heston | Fully fine-tuned | 0.160650 | **0.153222** | 0.160264 |
| Heston | Pretraining only | 0.169803 | 0.151809 | **0.149143** |
| Heston | Output fine-tuned | 0.169492 | 0.151640 | **0.150269** |
| Heston | Fully fine-tuned | 0.169652 | 0.152293 | **0.150447** |
| Market data | Market trained | **0.180479** | 0.183007 | 0.186248 |

*Note: All entries are premium-normalised MSE at one basis point on matched 1M liabilities; bold marks the lowest MSE within each source and training variant.*

The table rules out a simple "pooling always wins" account. Full sharing is best for every Lifted Heston variant, while the multi-task model is best for every Heston variant. Both outperform the corresponding

1M-specific policies. Market-only training reverses the ordering: its 1M-specific policy beats both shared alternatives. Additional maturity data are most valuable when combined with stochastic-volatility pre-training, not as a mechanical consequence of sample size.

Averaged across the Heston and Lifted Heston rows, the shared model improves on 1M-specific estimation by 0.013156 MSE and the multi-task model by 0.010216. Multi-task MSE is 0.002941 above the fully shared model on average. Maturity-specific output heads retain most of the benefit, but do not improve the average once the hidden representation is shared.

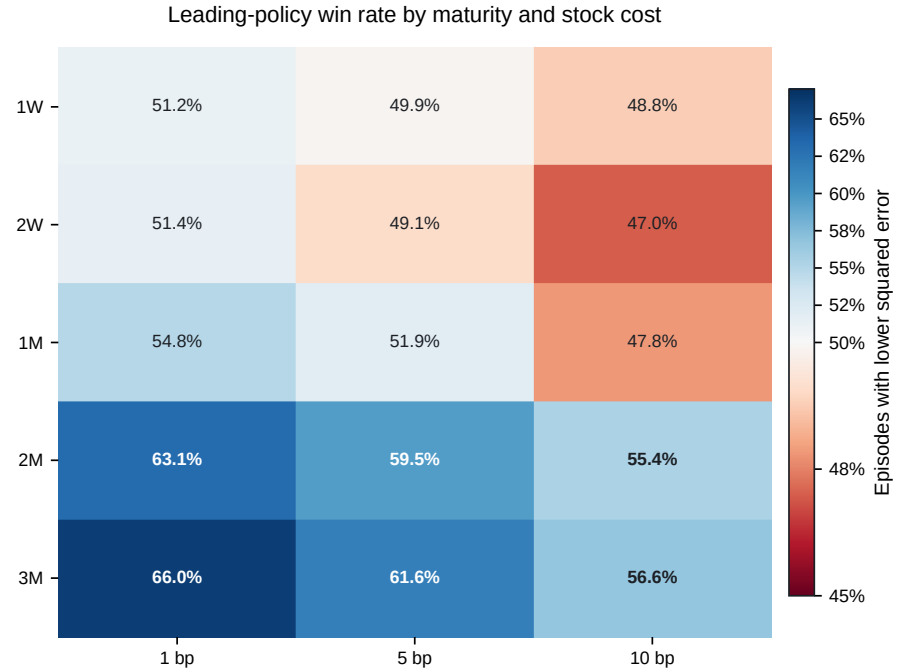



Figure 5.8. Share of matched episodes in which the shared Lifted Heston policy has lower squared terminal error than Black–Scholes delta, by entry maturity and stock cost.

Figure 5.8 shows two monotonic gradients. At one basis point, the win rate rises from 51.2% for 1W liabilities to 66.0% for 3M. It falls with stock cost at every maturity; at ten basis points the three shortest maturities drop below 50%, while 2M and 3M remain above 55%. More hedging opportunities strengthen the policy's advantage, and trading costs erode it.

# 6 Discussion

Three design choices explain the ranking. The loss function decides which outcomes count as improvement; simulation supplies a prior over dynamic inventory decisions; and parameter sharing controls how evidence is borrowed across maturities.

## 6.1 What the quadratic objective buys

As Equation (3.6) and Figure 5.2 show, MSE improves when the variance reduction overwhelms the smaller squared-bias increase. Buehler et al. (2019) formalise precisely this terminal-risk trade-off: the hedge improves the risk measure it was trained to minimise.

The decomposition in Figure 5.2 reveals the implicit risk preference: the policy reduces errors in the

worst 5% while slightly worsening the remaining 95%. MAE changes little because it weights errors linearly; under MSE, an error four times as large receives sixteen times the weight. Tail compression can thus improve MSE materially while barely moving MAE and lowering average P&L.

Zero-target MSE is direction-agnostic. A short straddle is exposed to move magnitude in either direction, and the loss penalises large gains as well as losses; it contains no term for market direction, drawdown, or investor wealth. Sell-off protection requires a one-sided tail loss, state-dependent utility, or an explicit capital constraint. CVaR concentrates weight on adverse outcomes (Rockafellar and Uryasev, 2000), while utility-based hedging prices the exchange between expected P&L and risk (Hodges and Neuberger, 1989).

The self-financing decomposition locates the cost in stock trading. The shared Lifted Heston and market-trained policies have similar turnover, yet the former records a much more negative stock-hedge contribution; commissions explain little of the gap. A short-gamma hedge tends to buy after rises and sell after falls. Changing the timing and size of those trades can prevent a large terminal mismatch while worsening average trading P&L. The network reallocates the risk of an existing option liability; it does not create the variance premium earned at entry.

Calendar concentration sharpens the interpretation. Almost all aggregate MSE improvement occurs in 2018 and 2020; 2023 records the largest shortfall. The policy behaves like state-contingent protection, paying off when benchmark errors become large and costing money when its extra intervention is unnecessary. MSE ranks that exchange; the value placed on tail risk and capital relief determines whether it is worth buying.

## 6.2 Simulation as a policy prior

Simulation supplies repeated hedging problems from which the policy learns the relation between observable state, inventory, and the next position. Historical option panels contain many overlapping observations but few independent volatility regimes; synthetic episodes broaden the state transitions without asking an eighteen-month window to estimate the policy from scratch. That data-scarcity rationale motivates Wiese et al. (2019) and the emphasis on generator robustness in Limmer and Horvath (2023).

Market fine-tuning may add little because simulation already contains transaction costs, prior inventory, and self-financing accounting. It can teach state-dependent trading logic resembling a no-trade policy before seeing market episodes. The rolling window can correct the state distribution, but supplies little independent evidence against a representation learned from far more episodes. Output-layer adaptation changes only the final map; the L2-SP penalty and lower full-network learning rate constrain broader changes. The design combines a strong simulated prior with a weak adaptation signal.

The shapes in Figure 6.1 match this mechanism. Simulation pretraining improves steadily across its training budget. Both fine-tuning routes adjust rapidly and then flatten, with little separation between them; market-only training improves more gradually and shows greater late-epoch variation. Most of the reusable state–inventory relation is learned in simulation, leaving the rolling market window to make comparatively small corrections.

This differs from Mikkilä and Kanniainen (2023), whose intraday setting supplies far denser historical transitions and uses a different reward. Direct market learning becomes more plausible as the number of

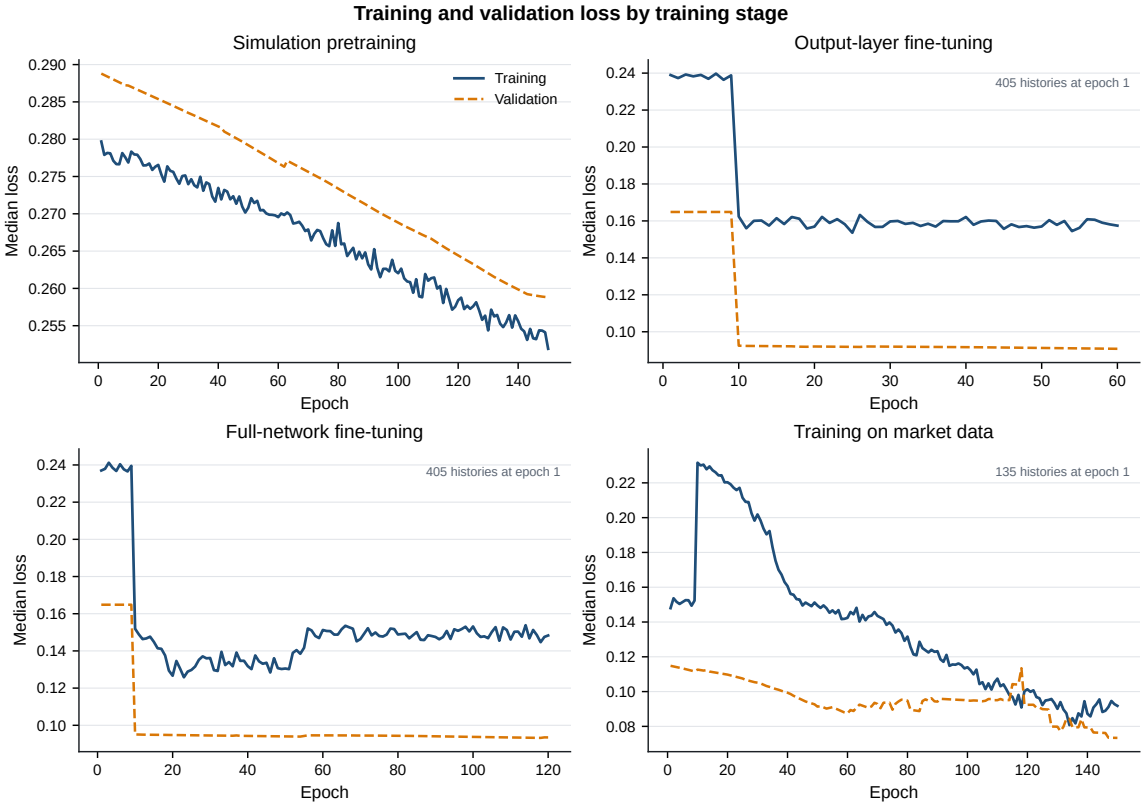



Figure 6.1. Median training and validation loss paths across simulation pretraining, output-layer fine-tuning, full-network fine-tuning, and market-only training. Panels use separate vertical scales.

informative and sufficiently independent decisions rises.

Training directly on market data nevertheless remains competitive, particularly at longer maturities in the descriptive results.[13] Simulation broadens the training distribution and can act as a prior over plausible dynamics; market data remain both a credible training source and the final test of whether the learned relation transfers. The weak fine-tuning effect says more about the marginal value of a short adaptation window than about the value of market evidence itself.

Figure 5.6 separates useful realism from generic fit. Black–Scholes is close on variables fixed by contract construction but misses ATM skew by more than five market interquartile ranges. Heston and Lifted Heston reproduce skew and term structure more closely, exposing their policies to states that resemble the SPX surface. All three miss variance levels and realised volatility. Richer dynamics matter when they improve decision-relevant inputs, not simply because they add latent factors.

Heston and Lifted Heston sharpen the point. Their observable heatmap rows are almost identical and their average hedging difference is small, despite Lifted Heston's additional volatility timescales. Those factors reach a stock-only policy through option prices, surface shape, and market histories. Abi Jaber and Li (2025) make the same task-dependent point: the best representation for fitting a surface need not be the best one for a particular control problem.

The conventional deltas provide a revealing negative comparison. Heston and Lifted Heston deltas perform worse than the simpler Black–Scholes and vendor deltas, even though their simulations support some of the strongest neural policies. The same model is a useful scenario generator and a poor local hedge

[13]The maturity-by-cost market-only comparison is reported in Appendix Table B.3 and Figure B.6.

calculator. Static calibration, local sensitivity accuracy, and terminal policy value are separate tests.

Simulation also creates a maintenance problem. A fixed generator encodes one calibration period and becomes stale as the market evolves. The concentrated calendar profile makes that risk concrete. A stronger design would mix generators and calibration dates, monitor observable drift, and adapt when the market leaves the training distribution. Adaptation should respond to measured distribution shift, not merely to the arrival of recent data.

## 6.3 Pooling as cross-maturity shrinkage

Parameter sharing acts as cross-maturity shrinkage. Maturities differ in gamma, theta, liquidity, and horizon but share the SPX underlying, volatility surface, inventory state, and transaction mechanism. Original maturity and current DTE allow a common network to condition on horizon rather than assume that contracts are identical. Sharing can therefore reduce estimation variance when a common state–hedge relation transfers across maturities, but creates negative transfer when maturity-specific effects dominate (Caruana, 1997).

The matched 1M reversal shows that this trade-off depends on training source. Stochastic-volatility pretraining favours pooling, whereas market-only training favours the 1M-specific model. The latter estimates a similar representation from roughly one fifth of the episodes available to pooled models, although within-year dependence limits the effective sample gain. This interaction suggests that simulation organises common inventory mechanics that pooling can estimate more precisely; without that prior, other maturities dilute the 1M signal. Maturity-specific heads outperform full sharing under Heston but not under Lifted Heston, and the simpler shared output performs best on average. Their additional flexibility could reduce bias in a larger sample, but here its estimation cost offsets the benefit. No architecture therefore dominates across training distributions.

The maturity gradient is consistent with a control-horizon mechanism. A 3M liability provides roughly twelve times as many daily hedge dates as a 1W liability, giving an inventory-aware policy more opportunities to use persistent surface information and the existing hedge. Weekly risk is concentrated in fewer moves that cannot be repaired after the event, while transaction costs penalise every additional intervention. The stronger 2M and 3M gains are therefore consistent with more opportunities for dynamic control, rather than merely lower average prediction error.

## 6.4 Limitations and next tests

The accounting identities reconcile numerically, and timing checks confirm that policies use only information available before execution.[14] The principal implementation limitation is optimisation failure in the Black–Scholes-pretrained models. Identical parameters across adaptation variants, zero effective gradients, and saturated outputs show that these networks did not train successfully. Their P&L remains arithmetically valid, but the rows cannot identify fine-tuning or pooling effects. Future experiments should pre-specify convergence criteria and redesign these models before computing factorial averages.

Second, the headline policy is selected from the evaluated specifications, while only nine annual periods

[14]The numerical evidence base is summarised in Appendix Table A.2; the concentration of optimisation failures is reported in Appendix Table B.2.

provide independent market evidence and most improvement occurs in 2018 and 2020. A future-period test should freeze the shared Lifted Heston, matched Heston, and market-trained policies alongside the vendor and Black–Scholes deltas. MSE decomposition, turnover, and P&L should also be fixed before evaluation.

Third, the experiment is end-of-day and stock-only: signals execute at the next close, the SPX close is not tradable, and overlapping episodes are not a funded account. A deployable test needs timestamped futures or SPY execution, basis, roll, funding, collateral, margin, and impact. Adding a volatility-linked hedge would test whether richer volatility dynamics matter when volatility risk is directly tradable.

Finally, the ranking is specific to a bounded delta-correction MLP and zero-target MSE. Comparing CVaR, utility, mean–variance, and capital objectives would test whether lower mean P&L is the price of tail compression or an artefact of symmetric loss. Rolling simulator mixtures with drift-triggered adaptation would address the remaining design question: when should the simulated prior be updated?

# 7 Conclusion

This dissertation examined how to hedge a short near-ATM SPX straddle when rebalancing is daily, costly, and based on information available at the decision time. A matched walk-forward experiment separated three issues often combined in deep-hedging studies: the distributional effect of the learned policy, the contribution of simulation, and the value of sharing information across maturities.

For RQ1, the lowest-MSE policy reduces premium-normalised MSE by 15.9% relative to vendor delta and 16.3% relative to Black–Scholes delta. Lower variance and fewer severe errors drive the gain; ordinary outcomes worsen slightly, mean P&L falls, and turnover rises. The policy provides tail-risk control at an observable economic cost. Zero-target MSE rewards the exchange because lower dispersion outweighs the increase in squared bias.

For RQ2, useful simulation transfer occurs during pretraining. Successfully trained Heston and Lifted Heston policies outperform market-only training on average, while subsequent output-layer and full-network adaptation barely changes MSE. Lifted Heston improves only slightly on Heston because additional latent dynamics matter through their observable price and surface implications. The weak stochastic-volatility deltas and strong simulation-pretrained policies also separate scenario generation from local hedge accuracy.

For RQ3, parameter sharing improves matched 1M performance on average across the successfully trained simulation-pretrained policies. Full sharing is strongest under Lifted Heston, multi-task sharing under Heston, and the 1M-specific model under market training. Cross-maturity shrinkage works when the shared representation transfers; sample size alone does not guarantee a gain.

Together, the findings establish a practical hierarchy for financial machine learning. The objective defines success, the training distribution determines transfer, and parameter sharing governs how scarce evidence is combined. Model complexity comes after those choices. The dissertation's contribution is a precise account of when a learned hedge reduces terminal risk, what that protection costs, and why the advantage appears.

# References

Abi Jaber, E. (2019) 'Lifting the Heston model', Quantitative Finance, 19(12), pp. 1995–2013.

Abi Jaber, E. and El Euch, O. (2019) 'Multi-factor approximation of rough volatility models', SIAM Journal on Financial Mathematics, 10(2), pp. 309–349.

Abi Jaber, E. and Li, S. (2025) 'Volatility models in practice: rough, path-dependent or Markovian?', Mathematical Finance, 35(4), pp. 796–817.

Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A. C., and Bellemare, M. (2021) 'Deep reinforcement learning at the edge of the statistical precipice', Advances in Neural Information Processing Systems, 34, pp. 29304–29320.

Bakshi, G. and Kapadia, N. (2003) 'Delta-hedged gains and the negative market volatility risk premium', Review of Financial Studies, 16(2), pp. 527–566.

Bayer, C., Friz, P., and Gatheral, J. (2016) 'Pricing under rough volatility', Quantitative Finance, 16(6), pp. 887–904.

Black, F. and Scholes, M. (1973) 'The pricing of options and corporate liabilities', Journal of Political Economy, 81(3), pp. 637–654.

Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019) 'Deep hedging', Quantitative Finance, 19(8), pp. 1271–1291.

Buehler, H., Murray, P., Pakkanen, M. S., and Wood, B. (2021) 'Deep hedging: learning to remove the drift under trading frictions with minimal equivalent near-martingale measures', arXiv:2111.07844.

Carr, P. and Wu, L. (2009) 'Variance risk premiums', Review of Financial Studies, 22(3), pp. 1311–1341.

Caruana, R. (1997) 'Multitask learning', Machine Learning, 28, pp. 41–75.

Cawley, G. C. and Talbot, N. L. C. (2010) 'On over-fitting in model selection and subsequent selection bias in performance evaluation', Journal of Machine Learning Research, 11, pp. 2079–2107.

Cboe Global Markets (2026) 'S&P 500 Index Options'. Available at: https://www.cboe.com/tradable-products/sp-500/spx-options/ (Accessed: 3 August 2026).

CRSP (2026) 'CRSP US Stock Databases'. Available at: https://www.crsp.org/research/crsp-us-stock-databases/ (Accessed: 17 August 2026).

Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018) 'Volatility is rough', Quantitative Finance, 18(6), pp. 933–949.

Heston, S. L. (1993) 'A closed-form solution for options with stochastic volatility with applications to bond and currency options', Review of Financial Studies, 6(2), pp. 327–343.

Hodges, S. D. and Neuberger, A. (1989) 'Optimal replication of contingent claims under transaction costs', Review of Futures Markets, 8, pp. 222–239.

Holm, S. (1979) 'A simple sequentially rejective multiple test procedure', Scandinavian Journal of Statistics, 6(2), pp. 65–70.

Leland, H. E. (1985) 'Option pricing and replication with transactions costs', Journal of Finance, 40(5), pp. 1283–1301.

Li, X., Grandvalet, Y., and Davoine, F. (2018) 'Explicit Inductive Bias for Transfer Learning with Con-

volutional Networks', Proceedings of the 35th International Conference on Machine Learning, PMLR 80, pp. 2825–2834.

Limmer, Y. and Horvath, B. (2023) 'Robust hedging GANs', arXiv:2307.02310.

Merton, R. C. (1973) 'Theory of rational option pricing', Bell Journal of Economics and Management Science, 4(1), pp. 141–183.

Mikkilä, O. and Kanniainen, J. (2023) 'Empirical deep hedging', Quantitative Finance, 23(1), pp. 111–122.

OptionMetrics (2026) 'IvyDB US: Historical Options and IV Data'. Available at: https://optionmetrics.com/united-states/ (Accessed: 14 August 2026).

Rockafellar, R. T. and Uryasev, S. (2000) 'Optimization of conditional value-at-risk', Journal of Risk, 2(3), pp. 21–42.

Whalley, A. E. and Wilmott, P. (1997) 'An asymptotic analysis of an optimal hedging model for option pricing with transaction costs', Mathematical Finance, 7(3), pp. 307–324.

Wiese, M., Bai, L., Wood, B., and Buehler, H. (2019) 'Deep hedging: learning to simulate equity option markets', arXiv:1911.01700.

# A Reproducibility details

This appendix records the architecture fixed before evaluation, sample coverage, optimisation, and accounting specification needed to reproduce the annual experiment with all five maturities represented in every test year. Changes to the underlying data files, features, test years, seeds, costs, or optimisation define a different experiment.

Table A.1. Reproducibility details

| Item | Value used in the study |
|---|---|
| Simulated training data | 336 episodes per simulation model and maturity group; 1,680 per model |
| Market inventory | 2,867 episodes; 620/605/604/539/499 by 1W/2W/1M/2M/3M |
| Out-of-sample evaluation years | Annual 2016–2024; nine test years containing all five maturities |
| Optimiser | AdamW; gradient clip 1; L2-SP $10^{-4}$ in full-network fine-tuning |
| Learning rates | $3 \times 10^{-4}$ simulation pretraining/training on market data alone; $3 \times 10^{-5}$ output layer; $1 \times 10^{-5}$ full-network fine-tuning |
| Batches | 1,024 simulation; 128 market; minimum four market steps |
| Numerics | Float32 differentiable training; float64 final accounting |

The values above define the experiment reported in the main text; code, configuration files, and complete exports should be retained with the submission archive.

## A.1 Sample and numerical checks

The main text reports only the features of the evaluation design needed to interpret the three research questions. Table A.2 records the complete sample and integrity summary, while Table A.3 gives the minimum episode requirements applied before model fitting.

Table A.2. Final evidence base and limits on interpretation

| Item | Evidence | Interpretive consequence |
|---|---|---|
| Out-of-sample evidence | 9 annual test years, 2016–2024 | Uncertainty is assessed across years rather than repeated rows within a year. |
| Option liabilities | 2,030 episodes on 462 entry dates | Daily hedge decisions within one episode are not treated as independent observations. |
| Models | 30 neural models, 5 seeds, and 5 maturities | The design covers several mechanisms, but only five optimisation runs are available for each model. |
| Accounting | 719,370 rows; max error $1.14 \times 10^{-11}$ | Terminal outcomes are numerically valid. |
| Training checks | 32 validation checks: 28 pass; 4 fail | Averages involving Black–Scholes neural pretraining and fine-tuning require sensitivity analysis. |
| Timing | No future rates, dividends, calibrations, or quotes | The results use only information available before each next-close execution. |

Table A.3. Sample inclusion rule and treatment of test years

| Set | Minimum episodes per maturity group | Role |
|---|---|---|
| Full market-data training window | 40 | Final fine-tuning/refit window |
| Initial training subset | 20 | Budget fitting before the validation sample inside each training window |
| Validation subset | 8 | Budget and no-trade-scale selection |
| Out-of-sample test year | 8 | Next annual outcome window |
| Test year with all five maturities | All four minima in all five maturity groups | Include the test year |
| Otherwise | Any maturity group below any minimum | Skip before fitting |

## A.2 Data coverage

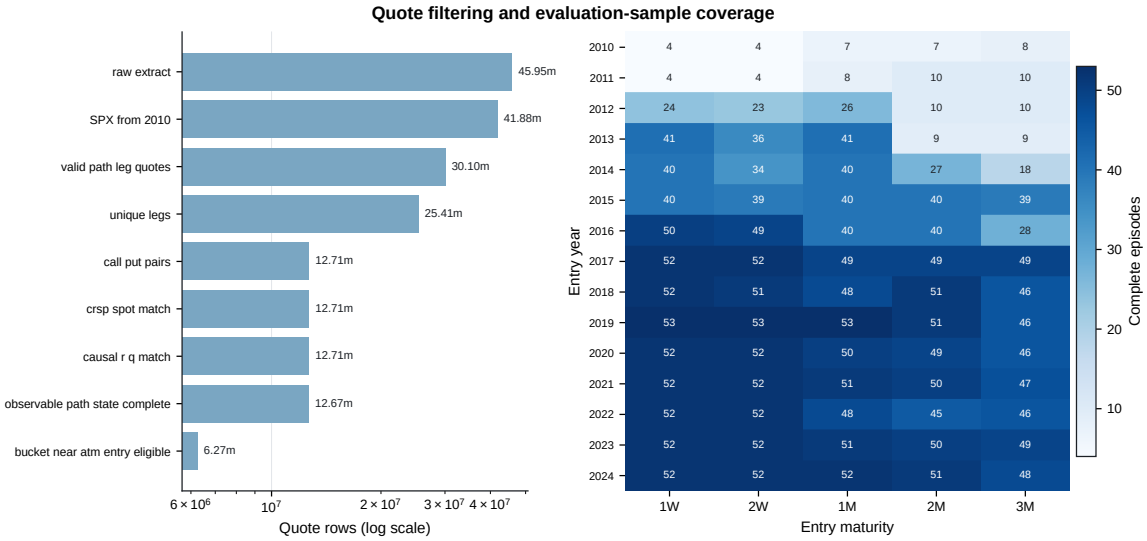



Figure A.1. Quote-funnel attrition and surviving episode coverage by year and maturity. Counts in the funnel are quote rows, not independent episodes.

The supplied extract contains approximately 45.9 million quote rows, of which about 6.27 million survive the near-ATM and entry-maturity eligibility rules before complete episodes are constructed. Figure A.1 also shows that the early sample contains few weekly episodes and that balanced coverage across all five matFurities becomes materially stronger from 2016. The annual balanced maturity coverage rule therefore removes 2012–2015 because the experiment cannot hold the maturity comparison fixed there, not because those years were selected after observing hedging performance.

## A.3 Calibration illustrations

The following figures document the upstream calibration pipeline that generated the stochastic-volatility states. They are retained as illustrations rather than evidence for the final model ranking. The final hedging experiment fixes $H = 0.35$, withholds hidden factors from the policy and evaluates transfer through observables; a visually attractive surface fit cannot by itself establish hedging value.

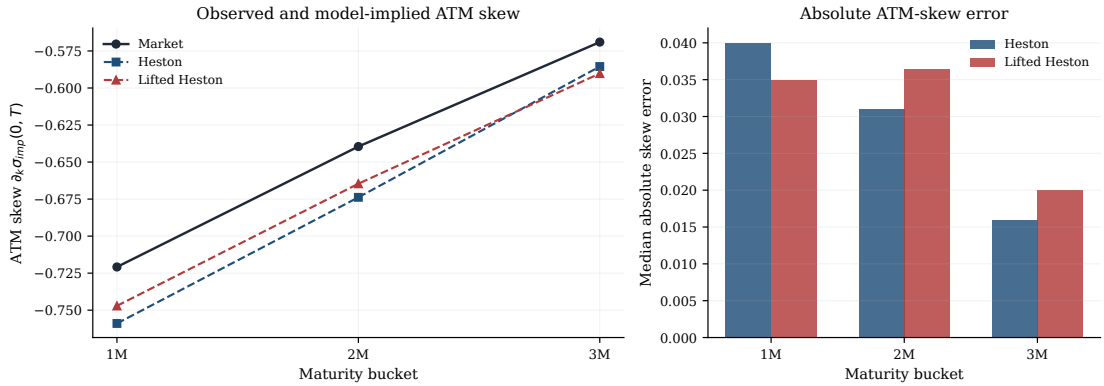



Figure A.2. ATM-skew fit by maturity on the common calibration target. Lifted Heston is closer at 1M, while Heston is closer at 2M and 3M.

Figure A.3 compares market, Heston, and Lifted Heston implied volatilities on six chronologically spaced calibration dates. For each date, the displayed contract set is the 3M maturity slice closest to 0.25 years with at least 12 common strikes. All common strikes in that slice are retained; the vertical line marks forward ATM. The calibration convention on the horizontal axis is $k = \log(K/F) = -m^F$, the negative of the forward-moneyness convention in Equation (4.1). Table A.4 reports the support and implied-volatility RMSE of the curves shown in the figure.

Table A.4. Six calibration-smile slices displayed in Figure A.3

| Date | DTE | Common strikes | Heston IV RMSE | Lifted IV RMSE |
|---|---|---|---|---|
| 4 Jan 2010 | 103 | 19 | 0.001609 | 0.001773 |
| 2 Aug 2011 | 109 | 15 | 0.001849 | 0.001679 |
| 19 Dec 2012 | 99 | 18 | 0.002494 | 0.004204 |
| 28 Oct 2014 | 94 | 25 | 0.003504 | 0.003757 |
| 9 Mar 2016 | 113 | 36 | 0.003351 | 0.003064 |
| 19 Aug 2020 | 77 | 46 | 0.002167 | 0.003401 |

*Note: Dates are chronological quantiles of the eligible calibration sample.*

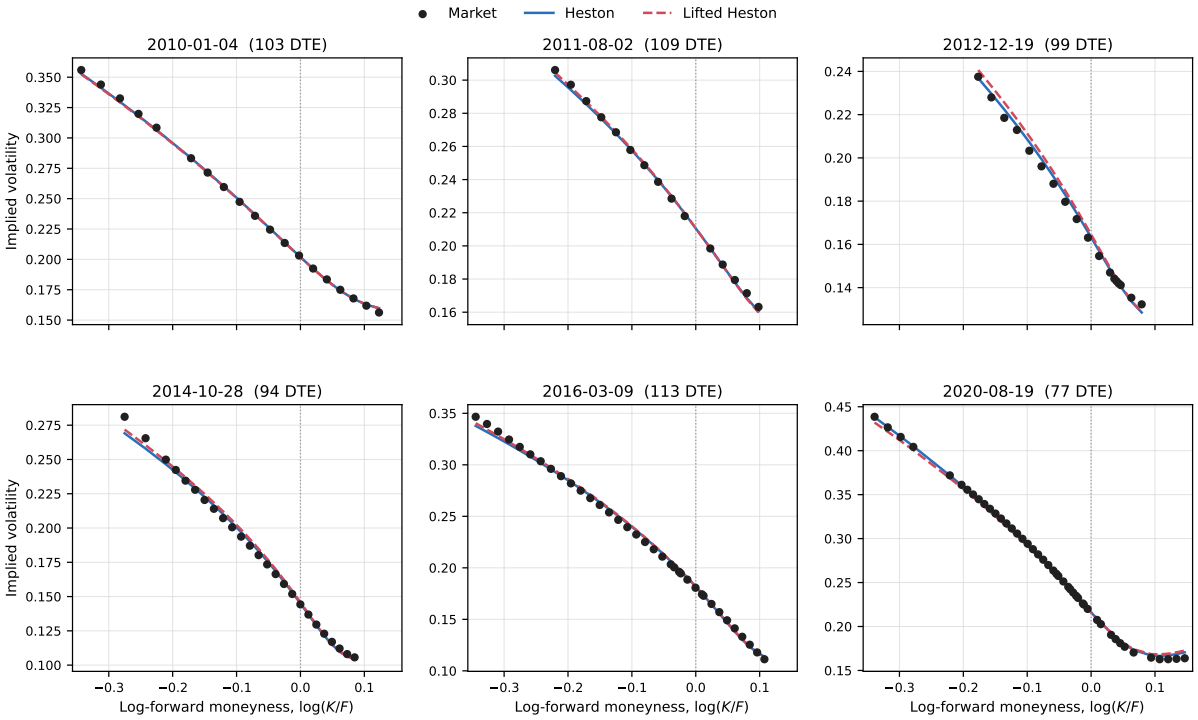



Figure A.3. Market, Heston, and Lifted Heston 3M implied-volatility smiles on six calibration dates. Each panel contains the complete common cross-strike slice; the dotted line marks forward ATM.

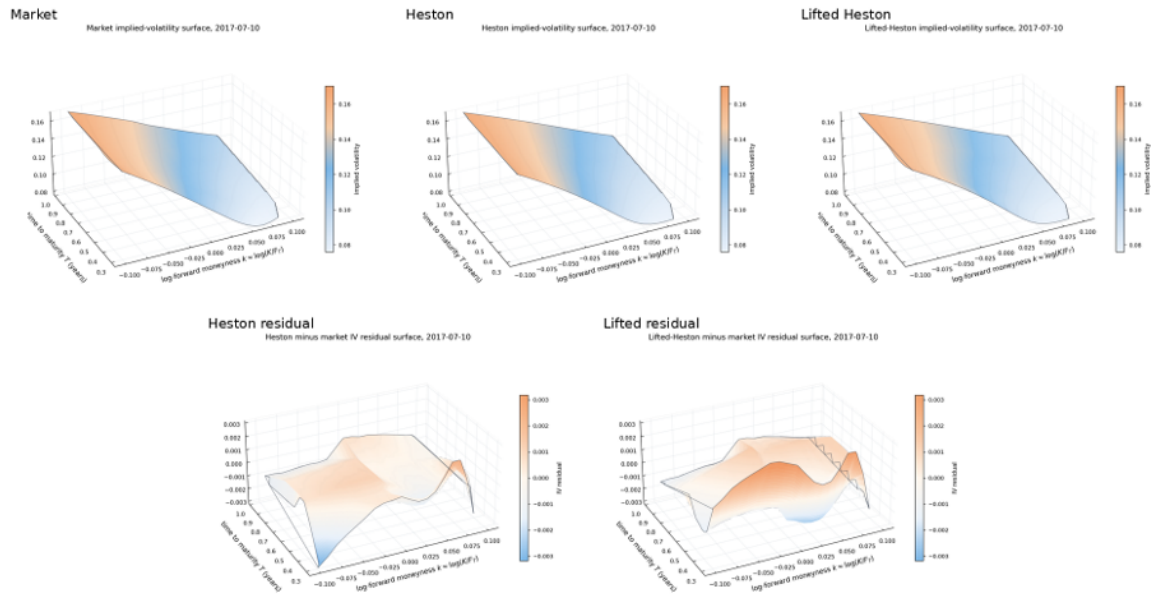



Figure A.4. Market, Heston, and Lifted Heston implied-volatility surfaces on 10 July 2017 (top), with signed Heston-minus-market and Lifted Heston-minus-market residual surfaces (bottom). All surfaces use common coordinates; the two residual panels share a narrower colour scale.

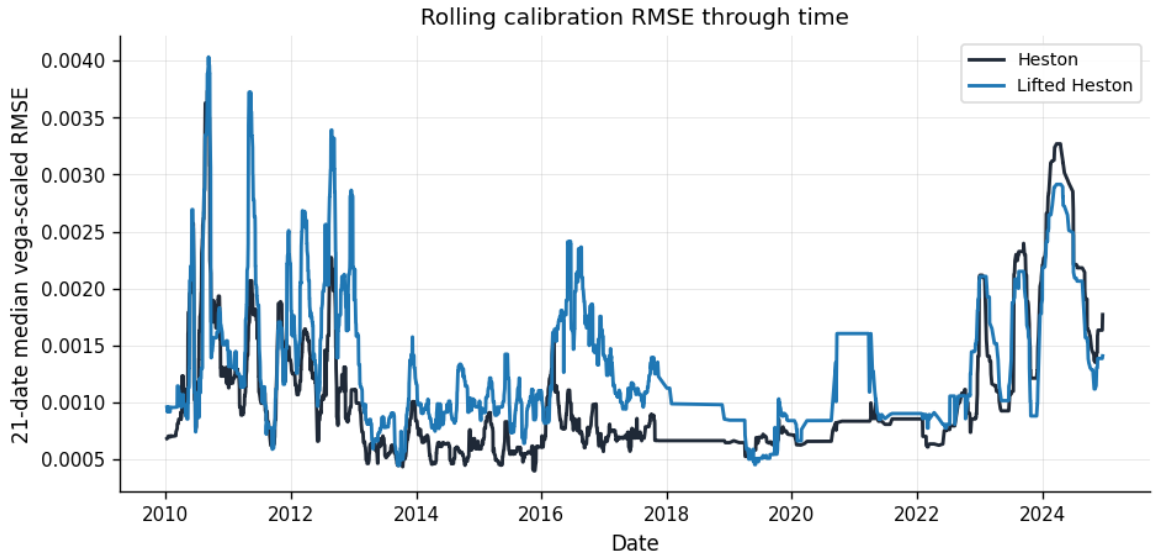



Figure A.5. Twenty-one-date rolling median of vega-scaled calibration RMSE for Heston and Lifted Heston.

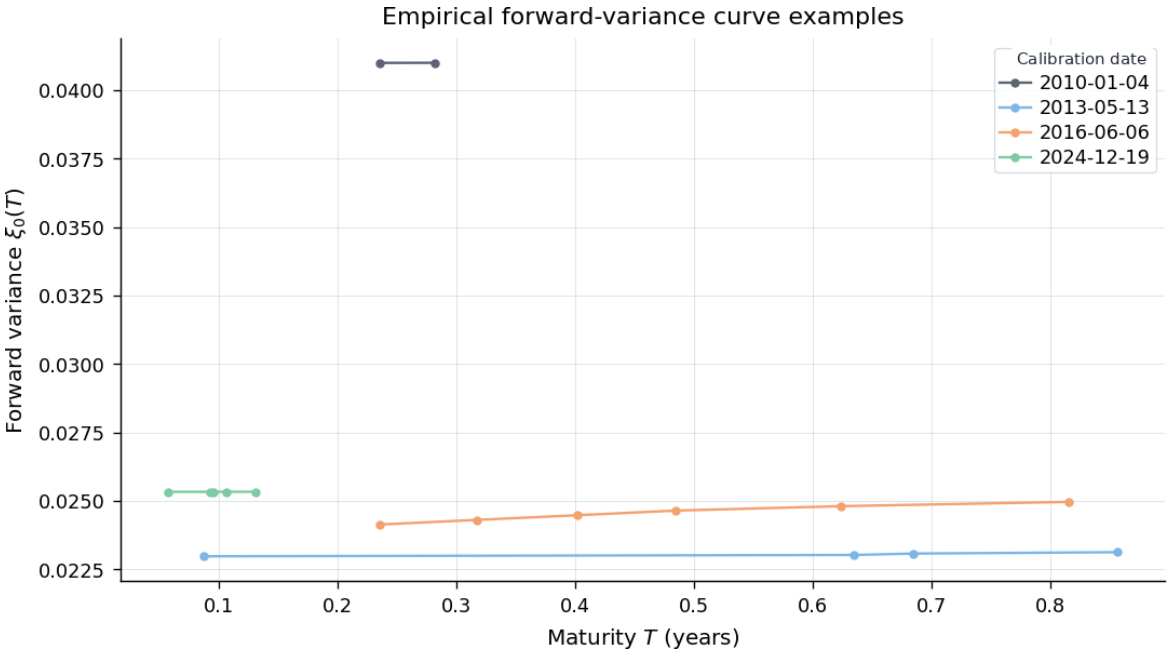



Figure A.6. Empirical forward-variance curves used as dated calibration inputs. Each line shows the regularised curve $\xi_0(T)$ across the available maturities for a representative calibration date.

## A.4 Network and inputs

The detailed architecture and information set are reported here because they are necessary for replication but are not themselves empirical findings.

Table A.5. Neural policy architectures and parameter sharing

| Design | Total parameters | Parameters used for one episode | Parameters updated in output-layer fine-tuning | Interpretation |
|---|---|---|---|---|
| One-month-specific model | 657 | 657 | 9 | Estimated using one-month options only |
| Shared across maturities | 657 | 657 | 9 | One set of parameters is used for all maturities, with current and original maturity supplied as inputs |
| Shared hidden layers with maturity-specific outputs | 693 | 657 | 45 total / 9 for the relevant maturity | Shared hidden layers with maturity-specific output layers |

The two hidden layers contain 648 parameters: $(31 \times 16 + 16) + (16 \times 8 + 8) = 648$. Each linear 8-to-1 output head adds nine parameters, so the multi-task model has $648 + 5 \times 9 = 693$ parameters in total and uses 657 for any one episode.

## A.5 Transaction-cost and chronology specification

Table A.6. Benchmark delta hedges and no-trade rules

| Target | Raw policy | No-trade policy | Inclusion rule |
|---|---|---|---|
| Black–Scholes | Rolled execution-time call-plus-put delta | State-dependent no-trade band | Primary classical benchmark |
| Vendor | OptionMetrics call-plus-put delta | State-dependent no-trade band | Observed-data benchmark |
| Heston | Delta from the most recent available calibration | State-dependent no-trade band | Included after checks of sign, units, maturity, and input construction |
| Lifted Heston | Delta from the most recent available calibration, checked by finite differences | State-dependent no-trade band | Included after numerical and interpolation checks |

The short call and put enter at their observed bids and close at observed asks. A fixed USD 4.50 round-trip fee equals 0.045 SPX index points at the contract multiplier of 100. Stock costs are proportional to traded notional at 1, 5, or 10 bp. Cash accrues financing and stock holdings receive dividend carry. Contract selection occurs after the signal close; option and initial hedge enter jointly at the next scheduled close; later hedge requests execute one close later; and the stock is liquidated with the option. Rates, dividends, and analytic calibrations are joined only from dates no later than the decision.

## A.6 Supporting derivations

This section supplies the intermediate algebra for two identities used in the main text.

### A.6.1 From Greek signs to the variance-spread approximation

Let $V_t = C_t + P_t$ be the value of a long straddle, with $\Delta_t^V = \Delta_t^C + \Delta_t^P$ and $\Gamma_t^V = \Gamma_t^C + \Gamma_t^P > 0$. Over a short hedge interval, Itô's formula gives

$$dV_t = \Theta_t^V dt + \Delta_t^V dS_t + \frac{1}{2}\Gamma_t^V S_t^2 \sigma_{\text{real},t}^2 dt. \tag{A.1}$$

The delta-hedged short position holds $\Delta_t^V$ index units and a cash balance $B_t = V_t - \Delta_t^V S_t$. Including dividend yield $q_t$, the financing rate $r_t$, and trading cost, its incremental value is

$$\begin{aligned} d\Pi_t^\Delta &= -dV_t + \Delta_t^V dS_t + q_t \Delta_t^V S_t dt + r_t B_t dt - dTC_t \\ &= \left[ -\Theta_t^V - \frac{1}{2}\Gamma_t^V S_t^2 \sigma_{\text{real},t}^2 + q_t \Delta_t^V S_t + r_t (V_t - \Delta_t^V S_t) \right] dt - dTC_t. \end{aligned} \tag{A.2}$$

At the locally observed implied volatility, the Black–Scholes–Merton equation is

$$\Theta_t^V + (r_t - q_t) S_t \Delta_t^V + \frac{1}{2}\Gamma_t^V S_t^2 \sigma_{\text{imp},t}^2 - r_t V_t = 0. \tag{A.3}$$

Substituting Equation (A.3) into Equation (A.2) cancels the financing and dividend terms:

$$d\Pi_t^\Delta = \frac{1}{2}(\Gamma_t^C + \Gamma_t^P) S_t^2 \left( \sigma_{\text{imp},t}^2 - \sigma_{\text{real},t}^2 \right) dt - dTC_t. \tag{A.4}$$

This is Equation (2.3). It is a local diffusion approximation: jumps and changes in the implied-volatility surface enter the residual rather than the displayed variance term.

### A.6.2 From the terminal loss to the MSE decomposition

Let $\mu = \mathbb{E}[Z]$. Adding and subtracting $\mu$ inside the quadratic loss gives

$$\begin{aligned} \mathbb{E}[Z^2] &= \mathbb{E}\big[((Z - \mu) + \mu)^2\big] \\ &= \mathbb{E}[(Z - \mu)^2] + 2\mu \mathbb{E}[Z - \mu] + \mu^2 \\ &= \text{Var}(Z) + \mu^2, \end{aligned} \tag{A.5}$$

because $\mathbb{E}[Z - \mu] = 0$. Hence, for matched policies $A$ and $B$,

$$\text{MSE}_A - \text{MSE}_B = (\text{Var}(Z_A) - \text{Var}(Z_B)) + \left( \mathbb{E}[Z_A]^2 - \mathbb{E}[Z_B]^2 \right). \tag{A.6}$$

The identity requires neither normality nor independence. It is the exact decomposition used to separate the variance and squared-bias contributions in Figure 5.2.

# B Additional robustness analyses

## B.1 Planned comparisons

The main chapter organises inference around three research questions. Table B.1 preserves all planned comparisons specified before final reporting so that the narrative reorganisation does not become selective reporting. Negative differences favour the first policy. The Holm adjustment controls all planned comparisons; the table should be read as a safeguard against selective reporting and overstatement, not as proof that economically small non-significant effects are exactly zero.

Table B.1. All comparisons planned before final reporting

| Dimension | Contrast | MSE diff. | 95% interval | Holm p |
|---|---|---|---|---|
| Fine-tuning | Output-layer fine-tuning - simulation pretraining only | -0.000356 | [-0.001619,0.000719] | 1.000 |
| Fine-tuning | Full-network fine-tuning - simulation pretraining only | -0.000251 | [-0.001916,0.001220] | 1.000 |
| Fine-tuning | Full-network fine-tuning - output-layer fine-tuning | +0.000104 | [-0.000279,0.000538] | 1.000 |
| Transfer | Simulation pretraining only - training on market data alone | +0.053986 | [0.029113,0.075406] | 0.0648 |
| Transfer | Output-layer fine-tuning - training on market data alone | +0.053630 | [0.029899,0.073709] | 0.0648 |
| Transfer | Full-network fine-tuning - training on market data alone | +0.053735 | [0.030632,0.073149] | 0.0648 |
| Simulation model | Heston - Black–Scholes | -0.201939 | [-0.224595,-0.182065] | 0.0330 |
| Simulation model | Lifted Heston - Heston | -0.005653 | [-0.016639,0.002602] | 1.000 |
| Parameter sharing (1M) | Multi-task model - shared across maturities | +0.050252 | [0.000175,0.118377] | 0.6759 |
| Parameter sharing (1M) | One-month-specific - shared across maturities | +0.063350 | [0.010630,0.129203] | 0.2567 |
| Parameter sharing (1M) | Multi-task model - one-month-specific | -0.013097 | [-0.019783,-0.008403] | 0.0330 |

The broad pretraining and Heston-versus-Black–Scholes comparisons include neural models that fail the training checks. The sensitivity analyses excluding failed neural fits in Sections 5.2 and 5.3 are therefore more useful for understanding why the result occurs, while this table remains the record of all comparisons planned before final reporting.

Twenty-eight of 32 validation checks pass. The four failures concern identical parameters after different fine-tuning stages, absent parameter and prediction movement, zero gradients, and bounded outputs fixed near their limits; they are learning failures rather than accounting failures. Accordingly, averages across simulation models, fine-tuning methods, and maturity-sharing designs that include Black–Scholes neural

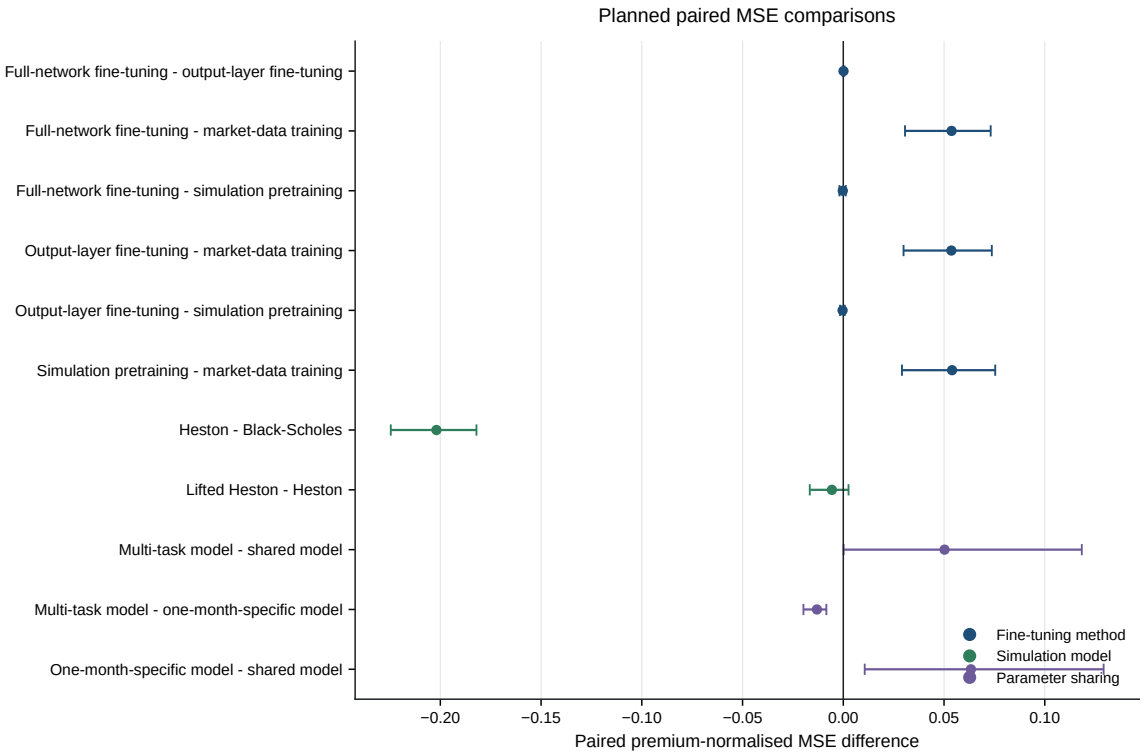



Figure B.1. All planned comparisons with unadjusted intervals from test-year resampling.

policies are accompanied by the sensitivity analyses excluding failed neural fits reported in the main text.

Table B.2. Concentration of neural training failures

| **Simulation model and maturity design** | **Fine-tuning comparisons** | **No effective parameter update** | **Models evaluated** | **Outputs fixed near their limits** |
|---|---|---|---|---|
| Black–Scholes, one-month-specific | 135 | 135 (100%) | 140 | 135 (96.4%) |
| Black–Scholes, shared across maturities | 135 | 135 (100%) | 140 | 135 (96.4%) |
| Black–Scholes, multi-task | 135 | 81 (60.0%) | 140 | 81 (57.9%) |
| Heston (all designs) | 405 | 0 | 420 | 0 |
| Lifted Heston (all designs) | 405 | 0 | 420 | 0 |
| Market data only (all maturity designs) | Not applicable | Not applicable | 135 | 0 |

Each fine-tuning comparison uses models with the same test year, simulation model, maturity-sharing design, and optimisation seed. The checks also cover simulation pretraining and training on market data alone.

## B.2 Temporal and regime stability

The main chapter retains the annual paired effect because calendar variation is central to inference. The following diagnostics provide supporting evidence without occupying the main narrative.

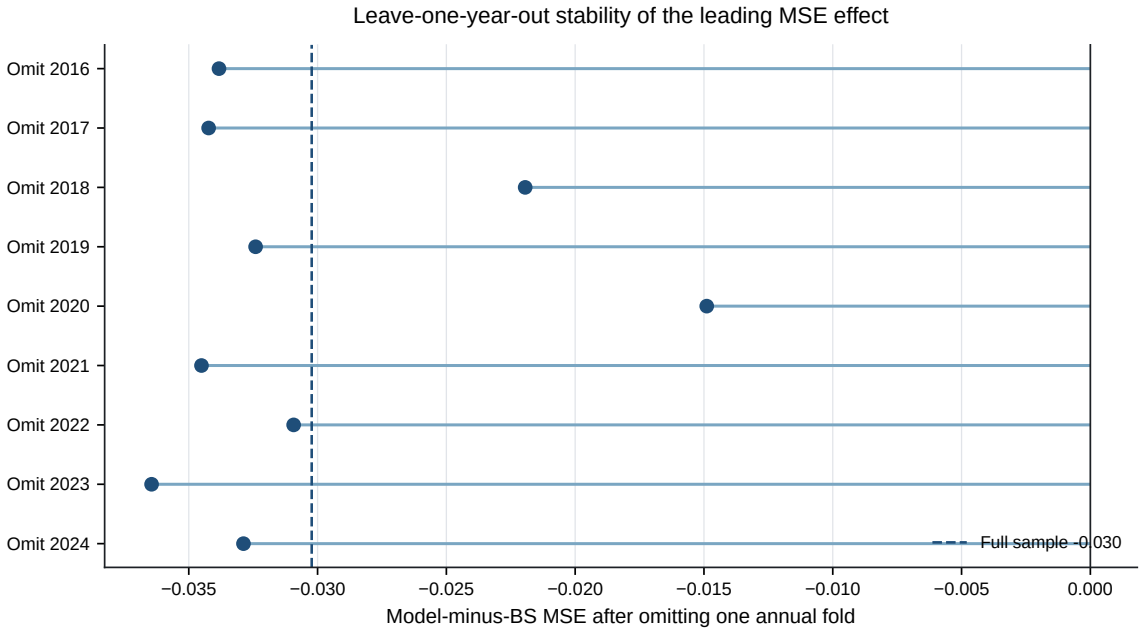



Figure B.2. Leave-one-year-out influence on the leading policy comparison.

The leave-one-year-out estimates remain negative under every omission, but the range is wide because only eight or nine years contribute to each estimate.

## B.3 Training and optimisation

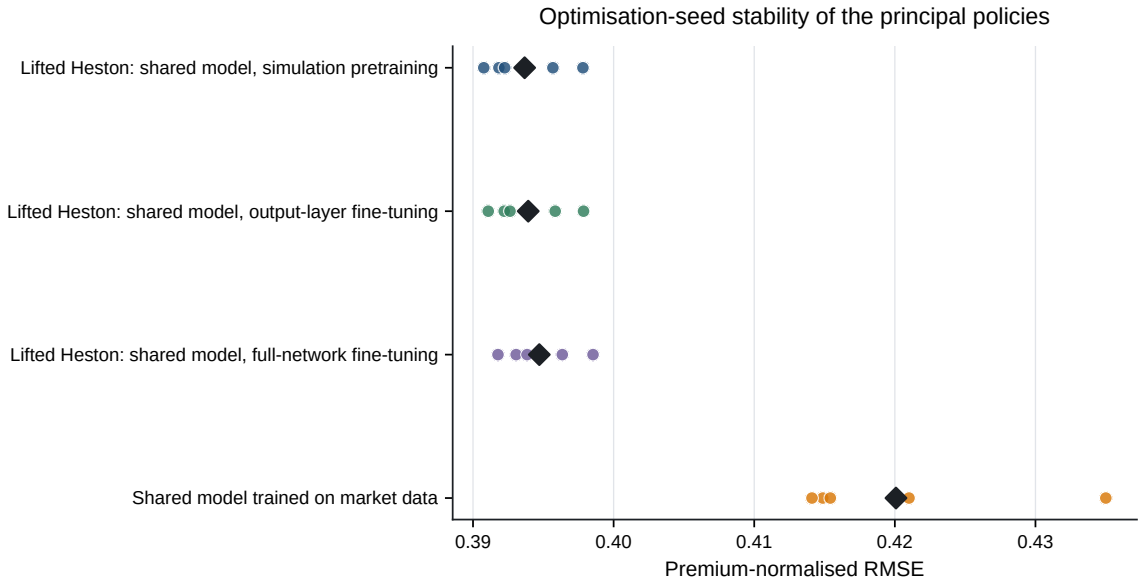



Figure B.3. Premium-normalised RMSE across five optimisation seeds for Lifted Heston training stages and market-only training.

Figure B.3 confirms that the leading result is not generated by one favourable initialisation. The three Lifted Heston training stages cluster tightly around RMSE 0.394–0.395, whereas the shared model trained

only on market data contains a visibly weak seed.

## B.4 No-trade benchmark

The candidate no-trade scale values are $\{0, 0.5, 1, 2, 4, 8\}$. Median Black–Scholes and vendor scales are zero at every cost, with many selections at the boundaries of the candidate set. Heston medians are 0, 0, and 1, while Lifted Heston medians are 0.5, 1, and 4 at 1, 5, and 10 bp. The pattern may reflect genuine cost trade-offs, a limited validation sample, or a coarse candidate set.




Figure B.4. Median validation-selected no-trade scale by benchmark and stock-cost level.

The figure protects against an over-strong interpretation of the no-trade benchmark. Zero or boundary selections do not show that no-trade theory is invalid; they show that this particular validation exercise often cannot distinguish a positive band from immediate rebalancing on the available validation sample.

## B.5 P&L decomposition

Figure B.5 reports the corresponding dollar differences, while main-text Figure 5.3 summarises the component contributions. For each option episode, terminal P&L is reconstructed as

$$\Pi_T = \Pi_T^{\text{option}} + \Pi_T^{\text{stock}} + \Pi_T^{\text{financing}} + \Pi_T^{\text{dividend}} - C_T^{\text{stock}} - C_T^{\text{option}}. \tag{B.1}$$

The accounting calculation verifies exact reconciliation to the reported terminal P&L. Because every policy hedges the same matched straddle episodes, the option-liability component is common. Differences arise primarily from the sequence of stock positions and, secondarily, financing, dividends, and trading costs.

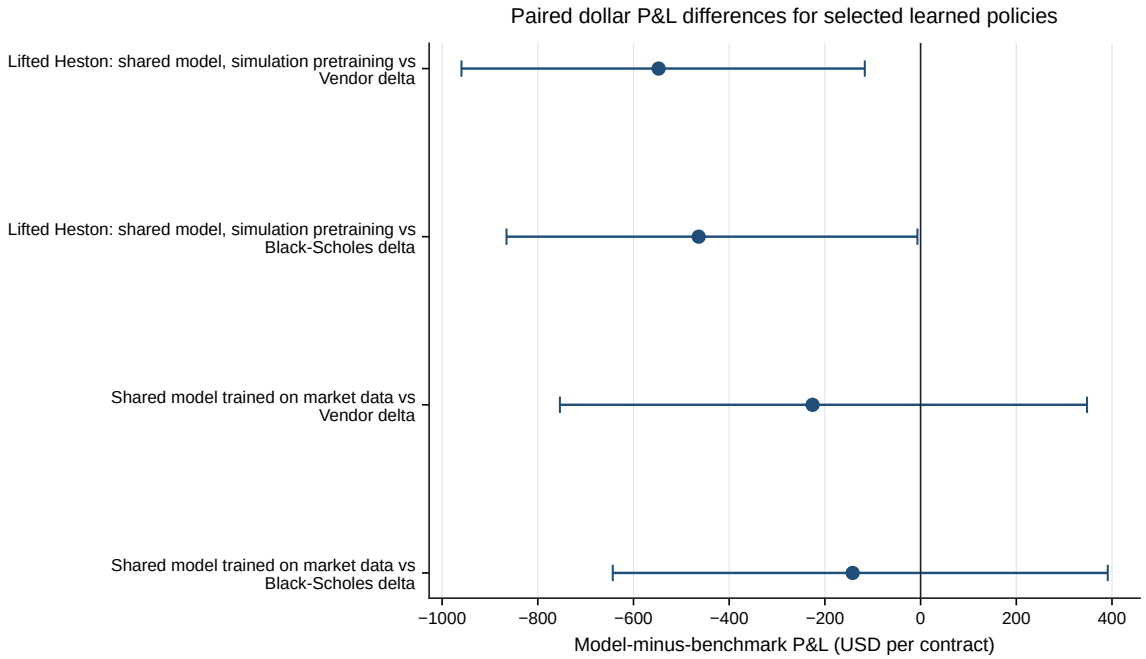



Figure B.5. Test-year-weighted dollar P&L differences relative to the Black–Scholes and vendor deltas; negative values favour the benchmark.

## B.6 Maturity, costs, and scale

Table B.3. MSE improvement from market-only training relative to Black–Scholes delta by original maturity

| **Stock cost** | **1W** | **2W** | **1M** | **2M** | **3M** |
|---|---|---|---|---|---|
| 1 bp | +0.6% | -0.8% | 0.0% | +11.7% | +21.2% |
| 5 bp | +0.2% | -1.6% | -0.8% | +9.7% | +20.1% |
| 10 bp | -0.5% | -2.4% | -1.8% | +6.9% | +17.6% |

Positive values favour training on market data alone. These averages are descriptive rather than a separately multiplicity-adjusted comparison.

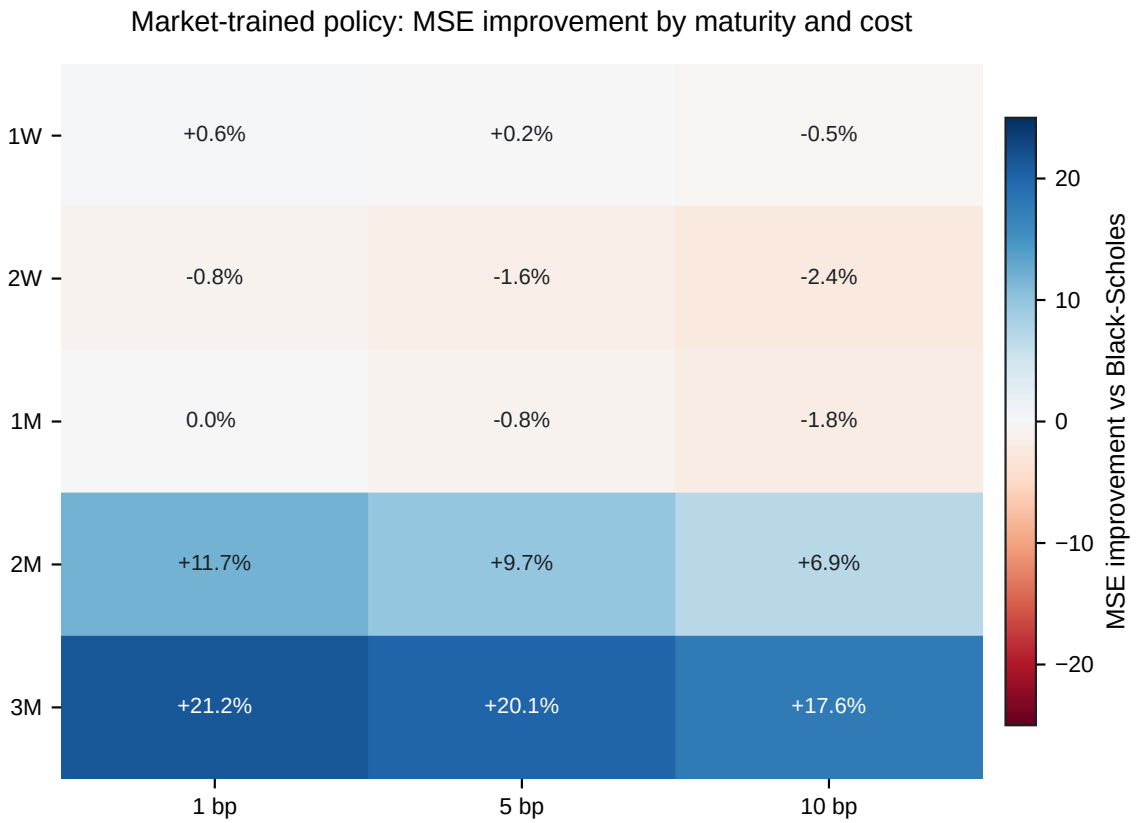



Figure B.6. Market-only MSE improvement relative to Black–Scholes delta by maturity and stock cost.

The walk-forward paths show that 2018 and 2020 are difficult environments for nearly every training approach, while the relative ranking varies materially by year. This is why the dissertation gives test years equal inferential weight rather than treating the combined number of repeated rows as the effective sample size.

Capital scale is illustrative rather than a funded backtest. Median entry premiums increase with maturity and one SPX contract carries index notional of several hundred thousand dollars. A USD 10,000 account cannot finance or margin an integer contract under the simple comparison, and margin, collateral remuneration, and portfolio netting are unavailable.

Figure B.8 makes the practical scale explicit. Treating each evaluated liability as one standard contract creates a large overlapping book, so the dollar results cannot be translated into a small-account return without margin rules, collateral remuneration, hedge-instrument basis, and portfolio netting.

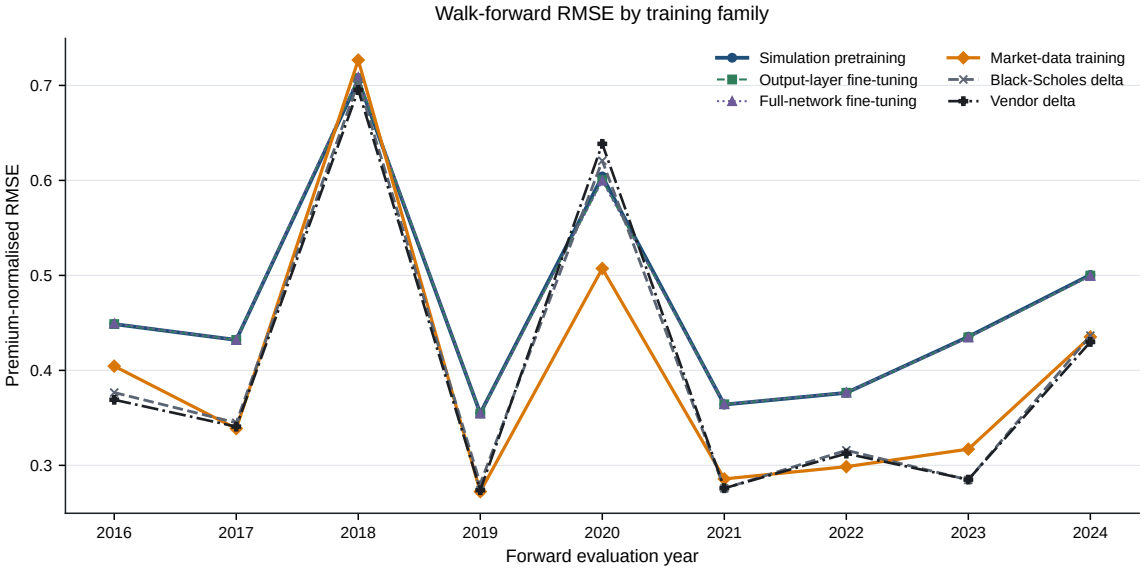



Figure B.7. Annual premium-normalised RMSE by training approach and delta benchmark.

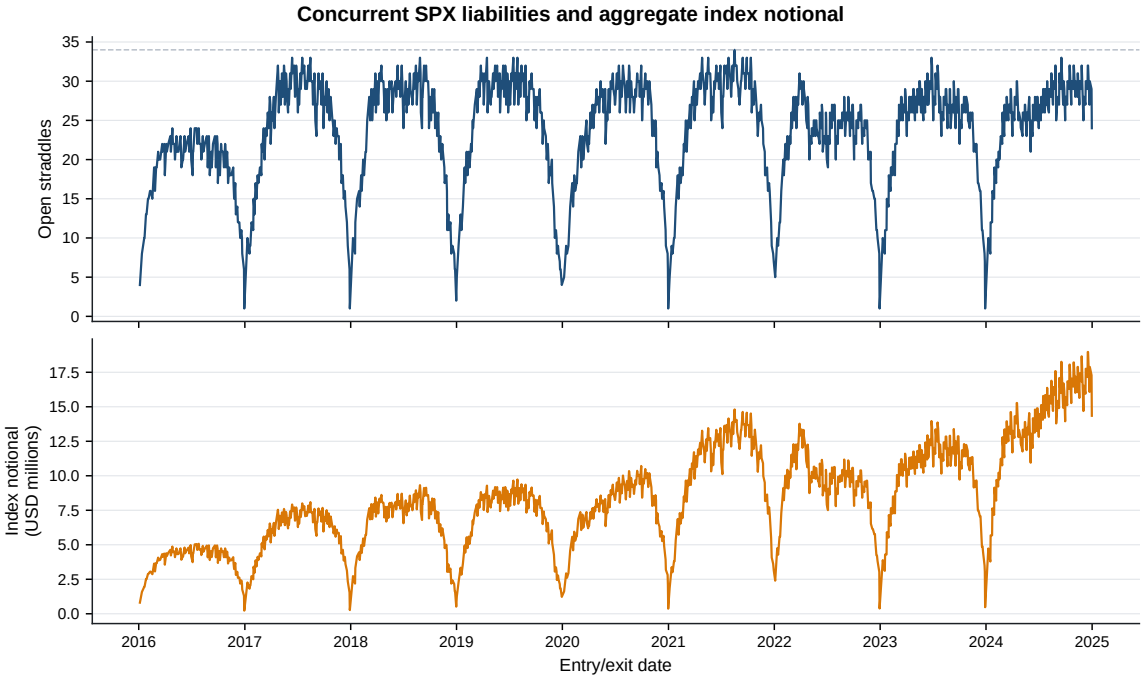



Figure B.8. Concurrent evaluated SPX liabilities and aggregate one-contract index notional through time.